Iterative Monotone Comparative Statics*

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Abstract

We propose a novel approach to comparative statics that applies to environments where complementarities play a critical role, including those in which existing methods for obtaining monotone comparative statics appear inadequate. Our approach is dynamic and, methodologically, in the spirit of the celebrated "correspondence principle" introduced in Samuelson (1947). It applies in particular to environments where : (a) adaptive dynamic adjustment processes are chaotic and may fail to converge, (b) equilibria are unstable, and (c) mixed shocks to the fundamental variables that nevertheless allow for unambiguous aggregate comparative statics.

Keywords: comparative statics; adaptive dynamic adjustment processes; monotone iterations; environments with complementarities

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1 Introduction

Comparative statics has always been a foundational tool in economic analysis. It examines how the set of optimal or equilibrium solutions of an economic model varies in response to changes in the model's parameters. Such predictions are important because they contain much of the empirical content of the economic model being studied.

The purpose of this paper is to propose a new approach to comparative statics in environments where complementarities play a critical role, including those in which existing methods for obtaining monotone comparative statics appear inadequate. Our results allow for comparative statics to be conducted in settings where there is a continuum of equilibria, where all equilibria are unstable, or in chaotic environments in which a variable of interest does not converge in the long run.

Our approach to comparative statics can be briefly described as follows. We first identify sharp (i.e., tight) bounds $\underline{a}(t)$ and $\overline{a}(t)$ for the asymptotic behavior of a large class of adaptive learning sequences $\left(a^k\right)_{k=0}^{\infty}$ of fundamental variables (e.g., the actions of players in games with strategic complementarities (GSC), or the prices of commodities in dynamic general equilibrium economies), starting from some initial point a^0. Given these bounds, we compare them (lower to lower and upper to upper) across different parameter values t. If the lower (upper) bound under a "new" parameter exceeds the lower (upper) bound under an "old" parameter, then the variables can be said to increase with the parameter change (in a weak sense). We also formulate alternative criteria by requiring that the lower bound under the new parameter exceeds the upper bound under the old parameter, or by requiring that the initial value a^0 be a steady state under the old parameter.

In addition, one need not compare the fundamental variables themselves, but only some statistics of interest derived from them. As we show, such statistics can be comparable in the long run, even when the underlying variables are not. In particular, we provide conditions under which comparative statics is conclusive for so-called "mixed shocks," i.e., shocks that affect some fundamental variables positively and others negatively.

Example 1. *Our leading example is a simple game with a continuum of actions and a continuum of equilibria. Analogous arguments apply to more complex coordination games with a finite number of actions and multiple equilibria.*

Consider the following joint venture. Players 1 and 2 choose actions a_1 *and* a_2 *from* $[0,1]$*, respectively, interpreted as their effort levels. The cost of taking action* a_i *for player* $i = 1,2$ *is* ca_i*, for some* $c \in (0,1)$*. The output of the team consisting of the two players is* $2\min\{a_1, a_2\}$*. Thus, the payoff of each player* i *is* $\min\{a_1, a_2\} - ca_i$*. This game has a continuum of equilibria: all pairs* (a_1, a_2) *such that* $a_1 = a_2$ *are equilibrium strategies.*

Suppose that players initially choose actions a_1^0 *and* a_2^0*, and the productivity of player* 1 *increases, so that the output becomes* $2\min\{ta_1, a_2\}$ *for some* $t > 1$*, and the payoffs are now* $\min\{ta_1, a_2\} - ca_i$ *for* $i = 1,2$*. Intuitively, the output should increase in response to this positive productivity shock.*

However, we cannot draw this conclusion by analysing best responses or comparing equilibria. Indeed, a change of parameter t ***is a "mixed shock" to the best response****: it increases the best response action of player 2, however, it* ***decreases*** *the action of player 1. Moreover, if initially* $a_1^0 = a_2^0 = a^0 \in (0,1)$*, then the total output in this equilibrium (for* $t = 1$*) is equal to* $2a^0$*. For* $t > 1$*,* ***the game has a continuum of equilibria****.* ***In some of them, output is higher than*** $2a^0$***, while in others it is lower than*** $2a^0$***.*** *Indeed, any effort profile* $(a/t, a)$*, with* $a \in [0,1]$*, constitutes an equilibrium strategy profile. In addi-*

tion, ***all equilibria are unstable*** *in the sense that, for some pairs of effort levels arbitrarily close to an equilibrium, the best-response dynamics does not converge to that equilibrium.*

Suppose that players adaptively learn to play this game with the new parameter t. In discrete time, this learning occurs through a sequence of action profiles $\left(a^k\right)_{k=0}^{\infty}$, starting from an initial action profile (a_1^0, a_2^0) (not necessarily an equilibrium). We will define the set $\mathcal{S}(a^0)$ of all "reasonable" (adaptive learning) sequences $\left(a^k\right)_{k=0}^{\infty}$, as well as action profiles (bounds) $\underline{a}(t)$ and $\overline{a}(t)$, such that in the long run (i.e., for large k), a^k cannot be "much smaller" than $\underline{a}(t)$ and cannot be "much larger" than $\overline{a}(t)$ for any sequence $\left(a^k\right)_{k=0}^{\infty} \in \mathcal{S}(a^0)$. For now, suppose that $\mathcal{S}(a^0)$ contains only the sequence of best responses to the opponents' actions from the previous period. Then

$$\underline{a}(t) = (\min\{a_1^0, a_2^0/t\}, \min\{a_1^0 t, a_2^0\})$$

and

$$\overline{a}(t) = (\max\{a_1^0, a_2^0/t\}, \max\{\min\{a_1^0 t, 1\}, a_2^0\}),$$

because these are the lowest and highest elements of $\mathbb{R}^2$ such that $\underline{a}(t) \leq a^k \leq \overline{a}(t)$ in the coordinate-wise ordering for all iterations a^k.

For this game, the bounds $\underline{a}(t)$ and $\overline{a}(t)$ for the best-response sequence $\left(a^k\right)_{k=0}^{\infty}$ will coincide with the bounds for other reasonable adaptive sequences; in addition, $br(\liminf_k a^k) = \liminf_k br(a^k)$ where br denotes the best response. We will prove in Theorem 1 that if this condition (and an analogous one for $\limsup$) is satisfied for all adaptive learning sequences from $\mathcal{S}(a^0)$, then both long-run bounds $\underline{a}(t)$ and $\overline{a}(t)$ are equilibria.

Suppose first that we start from (a_1^0, a_2^0) at $t = 1$. In the process of adaptive learning, after the change of t from $t = 1$ to $t > 1$, we end up between $\underline{a}(t)$ and $\overline{a}(t)$ for $t > 1$. Since the output at $\underline{a}(1)$ is no greater than that at $\underline{a}(t)$, and the output

at $\overline{a}(1)$ *is no greater than that at* $\overline{a}(t)$*, we claim that output should increase in response to this positive productivity shock, in a weak sense (see our Theorem 3).*

This conclusion becomes much stronger when $a_1^0 = a_2^0$ *is an equilibrium. In this case, the output at* a^0 *is equal to that at* $\underline{a}(t)$ *and is strictly smaller than that at* $\overline{a}(t)$ *for* $t > 1$ *(see our Theorem 4). Given the instability of equilibria, it is plausible to expect chaotic dynamics that may not converge to any equilibrium. We emphasize that our approach also applies to this case.*

Note that we reach the conclusion regarding the output despite the fact that the best response of one player (player 1) strictly decreases as t *increases.*[1] *Yet, the output in the long run is never lower. Thus,* ***applying our approach, we can conclude contrary to previous approaches the output will increase in response to an increase in*** t*.*

Our approach is iterative and methodologically in the spirit of the celebrated "correspondence principle," first presented in the work of Samuelson (1947) and later extended, most notably, in a series of papers by Echenique (e.g., Echenique (2002, 2004); see also McLennan (2015)). However, our approach departs from the existing literature in some important dimensions that we discuss in a moment.

Related Literature Performing comparative statics analysis on equilibrium problems is complicated. Especially in economic models with multiple equilibria, fixed-point comparative statics typically involves the tools of transversality and degree theory, and others from differential topology. These tools typically provide only weak local equilibrium comparative statics results, and even for these, they require stronger regularity conditions on the primitives than those in the context of implicit function based comparative statics of optimization problems. Alternatively, there is an extensive literature on fixed-point comparative statics for parameterized

[1] That is, the example violates an assumption that is typically made in the existing results on monotone comparative statics.

monotone operators and correspondences that transform chain-complete partially ordered sets.[2] A general limitation of these existing order-theoretic approaches is that they typically provide limited comparative statics information in the presence of multiple equilibria. That is, the comparative statics results pertain typically to only extremal equilibria (i.e., least/minimal or greatest/maximal) and the constructive nature of the comparative statics result does not hold for iterations from any initial point.[3]

A well-known approach to studying the equilibrium comparative statics of any equilibria is embodied in the so-called "correspondence principle," which was suggested originally in the seminal work of Samuelson (1947).[4] Here, one seeks to identify regularity conditions of optimization problems or equilibrium problems for unambiguous equilibrium comparative statics by refining away unstable equilibria, and then restricting attention to regular (or smooth) equilibria. This approach is inherently dynamic, and can be applied when equilibria are locally unique and amenable to applications of the implicit function theorem. Echenique (2002) has extended these ideas substantially, and has been able to prove stronger versions of the correspondence principle for GSC on complete lattices A when there is a convex set of parameters T. For example, Echenique (2002) showed that in GSC's, a continuous equilibrium selector $t \to a^*(t)$ is strictly increasing if and only if it

[2] Fixed-point comparative statics results for strong set order monotone (or "ascending") correspondences in complete lattices can be found in Topkis (1998), Chapter 2, Section 5. See also Theorem 2 in Chapter 10 in Veinott (1992). For comparative statics results on extremal fixed points, see Milgrom and Roberts (1990) Theorem 6 and Milgrom and Shannon (1994) Theorem 13. Sabarwal (2025a) Theorem 3 provides results on the existence of order nearest comparative statics for subsets of equilibria, but does not discuss computable approaches to characterizing them. We discuss relations of his results to ours in Section 4.

[3] See, for example, the computable comparative statics results for Nash equilibrium in interim Bayesian supermodular games in Van Zandt (2010). Also, see the discussion in Balbus et al. (2022). There, iterations need to start from least (resp., greatest) elements of the domain.

[4] See also McLennan (2015) for an interesting recent discussion of the correspondence principle, citations of the extensive literature and implications for equilibrium comparative statics.

selects stable equilibria.[5]

Our paper shares with the correspondence principle the idea that the identification of monotone comparative statics is critically tied to a dynamic approach. For example, if an equilibrium at the original set of parameters is locally stable, then one can develop sufficient conditions on the behavior of this dynamical system that guarantee that starting from the equilibrium for the old parameter, the dynamical system will actually converge to the new equilibrium for small changes of the parameter.

But this leaves open many new interesting questions. First, what happens if, in contrast to the existing related papers, we allow an initially studied point to be *not comparable* to its image under studied best response / law of motion? Second, what if the domain transformed by our monotone mapping is not a complete lattice, but only a sigma-complete? Third, when can a conclusive comparative static regarding ordered changes in some statistics or aggregates be performed, even though ordered changes in the fundamental variables defining those statistics or aggregates themselves are ambiguous?

Although technical, these points turn out to be critical when the setting under study is chaotic and the observed outcomes do not converge to any equilibrium. Indeed, when the initially studied point is not comparable to its image, even a monotone adjustment process does not guarantee convergence, and the resulting dynamics can become chaotic. Moreover, if a lattice is not complete, one cannot invoke a transfinite induction argument to guarantee the existence of accumulation points (see, e.g., Balbus et al. (2025); Echenique (2005)). Our results also prove useful when the equilibria under study are unstable or when there is a continuum of equilibria. To the best of our knowledge, our results concerning comparative

[5] See also Echenique (2002, 2004) for the precise formulations of various versions of this result. Notice, in the presence of multiple equilibria, the existence of continuous equilibrium selector is an added complication in applying Echenique (2002) results.

statics for mixed shocks have not been available in the literature so far.

We answer the above questions in the reminder of the paper. In the next section, we define mathematical terminology. In Section 3, we introduce our main concepts. Our main results and relation of our approach to the existing results are in Section 4. The new results on mixed shocks to fundamental variables and comparative statics of their statistics or aggregates are in Section 5. In Section 6, we present a series of applications. We include the proofs into the main text, delegating only the proofs of more technical results to Appendix.

2 Preliminaries

We start with introducing some basic definitions. A partially ordered set (or *poset*) is set A equipped with a partial order $\geq$. For $a', a \in A$, we say a' is *strictly higher* than a, and write $a' > a$, whenever $a' \geq a$ and $a' \neq a$. A poset $(A, \geq)$ is a *lattice* if for any $a, a' \in A$ there exists the *join* $a \vee a' := \sup\{a, a'\} \in A$ and there exists the *meet* $a \wedge a' := \inf\{a, a'\} \in A$. A lattice A is *complete (resp. sigma-complete)* if there exists $\bigvee B := \sup B \in A$ and $\bigwedge B := \inf B \in A$, for any (resp. any countable) $B \subseteq A$. A subset $B \subset A$ is a *sublattice* of A if B is a lattice in the order induced from A, i.e. the join $a \vee a'$ and the meet $a \wedge a'$ as defined in $(A, \geq)$ belong to B for all $a, a' \in B$.

Let $(A, \geq)$ and $(B, \geq)$ be posets. A mapping $f : A \to B$ is order-preserving (or increasing) on A if $a' \geq a$ implies $f(a') \geq f(a)$ for a, a' in A. Let $F : A \rightrightarrows B$ be a nonempty-valued correspondence. Assume that for each $a \in A$ the set $F(a)$ has the greatest and the least elements in A, and these elements belong to $F(a)$. Denote them with $\overline{F}(a) := \sup F(a)$ and $\underline{F}(a) := \inf F(a)$. We say F is *weakly increasing*[6] whenever $a' > a$ implies that $\overline{F}(a') \geq \overline{F}(a)$ and $\underline{F}(a') \geq \underline{F}(a)$. We say

[6] Observe that weak monotonicity of a correspondence is a weaker notion than (Veinott-) strong set-order monotonicity. In particular, F need not be sublattice valued.

F is *strongly increasing* whenever $a' > a$ implies that $\underline{F}(a') \geq \overline{F}(a)$.

A sequence $(a^k)_{k=0}^{\infty}$ of elements of A is *increasing* if $a^{k+1} \geq a^k$ for each k. It is *strictly increasing* if $a^{k+1} > a^k$ for each k. *Decreasing* and *strictly decreasing* sequences are defined in the obvious dual manner. A *monotone sequence* then is either increasing or decreasing. We say that an increasing (resp., decreasing) sequence $(a^k)_{k=0}^{\infty}$ *converges* to $a \in A$ whenever $\bigvee_{k\geq 0} a^k = a$ (resp., $\bigwedge_{k\geq 0} a^k = a$). That is, when a is the supremum (resp., infimum) of the increasing (resp., decreasing) sequence.

Suppose A and B are sigma-complete lattices. The mapping $f : A \to B$ is *upward order continuous* (resp., *downward order continuous*) if for any increasing (resp., decreasing) sequence $(a^k)_{k=0}^{\infty}$ with $a^k \in A$, we have: $f\left(\bigvee_{k\geq 0} a^k\right) = \bigvee_{k\geq 0} f(a^k)$ $\left(\text{respectively } f\left(\bigwedge_{k\geq 0} a^k\right) = \bigwedge_{k\geq 0} f(a^k)\right)$. The mapping f is then *order continuous* if it is both upward and downward order continuous. Notice, if f is upward or downward order continuous, it is order-preserving or increasing mapping on A.[7] Clearly, a weakly increasing correspondence that is single valued is an increasing mapping.

3 Setting and concepts

We study an economy in which some variables or actions of the economy agents, which we call fundamental, are described by an element a of A, interpreted as the current state. A mapping f (or a correspondence F) is a reaction function. For any a, $f(a)$ or $F(a)$ describes the way in which a would change from the current state in one period if *every agent would optimally respond to the current state* a. If $a = f(a)$ (or a belongs to $F(a)$), then the economy is in equilibrium.

[7] If a mapping is upward (resp., downward) order continuous, it is also by definition sup (resp, inf) preserving. So our definitions here coincide with standard definitions of order continuity (e.g., Dugundji and Granas (1982), p 15).

3.1 Adaptive dynamic adjustment

The idea of our approach to comparative statics can be described as follows: suppose we initially observe a state $a^0 \in A$. This a^0 need not even be an equilibrium. Then an exogenous parameter t of the setting changes, and this initiates an adaptive dynamic adjustment process in the setting with the new parameter. Assume that this dynamic process must have the form of some sequence $(a^k)_{k=0}^{\infty}$ from a class of sequences $\mathcal{S}(a^0)$. We will focus on the following, large class of sequences $\mathcal{S}(a^0)$.

We consider a lattice A that is sigma-complete. Let $F : A \rightrightarrows A$ be a weakly increasing correspondence. A sequence $(a^k)_{k=0}^{\infty}$ starting from a^0 is an *adaptive sequence* if

$$\exists_{\gamma \in \mathbb{N}} \, \forall_{k \in \mathbb{N}} \, \underline{F}(\inf\{a^k, ..., a^{k-\gamma+1}\}) \leq a^{k+1} \leq \overline{F}(\sup\{a^k, ..., a^{k-\gamma+1}\}),$$

where it is assumed $a^{k-\gamma+1} = a^0$ for $k < \gamma - 1$. Denote by $\mathcal{S}(a^0)$ the set of all adaptive sequences starting from a^0.

The idea is to be quite agnostic about the specific form of adjustments. We basically impose one postulate, namely, that agents respond to some finite history of variables or actions observed in previous periods. The parameter γ measures how far agents look into the past, and $\inf\{a^k, ..., a^{k-\gamma+1}\}$ and $\sup\{a^k, ..., a^{k-\gamma+1}\}$ are the extreme statistics of the variables or actions they observed in the past they look into. The correspondence $F : A \rightrightarrows A$ represents possible responses if *every agent would optimally respond to the current state* a.

Echenique (2002) in his analysis of GSC also used sequences $(a^k)_{k=0}^{\infty} \in \mathcal{S}(a^0)$, however, only under the additional assumption that $a_0 \leq \underline{F}(a^0)$. This assumption allowed him to obtain *equilibrium* bounds for accumulation points of sequences in $\mathcal{S}(a^0)$ for arbitrary length of the memory (parameter γ). In our case, when $a_0 \not\leq \underline{F}(a^0)$, accumulation points of sequences in $\mathcal{S}(a^0)$ can depend on γ. In the

next subsection we show how one can obtain bounds for all sequences in $\mathcal{S}(a^0)$ irrespectively of the history γ.

3.2 Bounds of adaptive sequences

At first sight, studying all sequences $(a^k)_{k=0}^{\infty} \in \mathcal{S}(a^0)$ seems intractable. However, we will show next that it reduces to studying their lower and upper bounds $\underline{a}$ and $\overline{a}$. To define these bounds, we first define by induction sequence $\underline{a}^{k,\gamma}$ (and simultaneously, sequence $\overline{a}^{k,\gamma}$). Let $\underline{a}^{0,\gamma} = \overline{a}^{0,\gamma} = a^0$ for all $\gamma \in \mathbb{N}$, and let $\underline{a}^{k+1,\gamma} = \underline{F}(\inf\{\underline{a}^{k,\gamma}, ..., \underline{a}^{k-\gamma+1,\gamma}\})$ and $\overline{a}^{k+1,\gamma} = \overline{F}(\sup\{\overline{a}^{k,\gamma}, ..., \overline{a}^{k-\gamma+1,\gamma}\})$. It is assumed that $\underline{a}^{k-l,\gamma} = \underline{a}^{0,\gamma}$ and $\overline{a}^{k-l,\gamma} = \overline{a}^{0,\gamma}$ for $l > k$. Next, for any given γ, define $\liminf_k \underline{a}^{k,\gamma} = \bigvee_k \bigwedge_{l \geq k} \underline{a}^{l,\gamma}$ and $\limsup_k \overline{a}^{k,\gamma} = \bigwedge_k \bigvee_{l \geq k} \overline{a}^{l,\gamma}$. Observe that $\liminf_k \underline{a}^{k,\gamma}$ and $\limsup_k \overline{a}^{k,\gamma}$ exist by sigma-completeness of the lattice.

Then, for each adaptive sequence $(a^k)_{k=0}^{\infty}$, there exists a γ such that $\underline{a}^{k,\gamma} \leq a^k \leq \overline{a}^{k,\gamma}$. This implies that

$$\liminf \underline{a}^{k,\gamma} \leq \liminf a^k \leq \limsup a^k \leq \limsup \overline{a}^{k,\gamma}.$$

In addition, the sequence $\left(\liminf_k \underline{a}^{k,\gamma}\right)_{\gamma=0}^{\infty}$ is decreasing while $\left(\limsup_k \overline{a}^{k,\gamma}\right)_{\gamma=0}^{\infty}$ is increasing. Let $\underline{a} = \lim_{\gamma\to\infty} \liminf_k \underline{a}^{k,\gamma}$ and $\overline{a} = \lim_{\gamma\to\infty} \limsup_k \overline{a}^{k,\gamma}$.

By the monotonicity of sequences $\left(\liminf_k \underline{a}^{k,\gamma}\right)_{\gamma=0}^{\infty}$ and $\left(\limsup_k \overline{a}^{k,\gamma}\right)_{\gamma=0}^{\infty}$, we obtain: $\underline{a} \leq \liminf a^k \leq \limsup a^k \leq \overline{a}$ for any adaptive sequence $(a^k)_{k=0}^{\infty}$. Formally, this does not mean that $\underline{a} \leq a^k \leq \overline{a}$ for remote elements of any adaptive sequence $(a^k)_{k=0}^{\infty}$. However, this captures the intuition that for remote elements of any adaptive sequence, a^k is not much lower than $\underline{a}$, and not much greater than $\overline{a}$.

Our bounds are sharp in the following sense. We say $\overline{b}$ and $\underline{b}$ are *bounds* of sequences in $S(a^0)$ if and only if for every sequence $(a^k)_{k=0}^{\infty}$ from $S(a^0)$ we have $\underline{b} \leq \liminf a^k \leq \limsup a^k \leq \overline{b}$. We say bounds $\overline{b}^*$ and $\underline{b}^*$ are *sharp (or tight)* for

$S(a^0)$ if for all bounds $\overline{b}$ and $\underline{b}$ we have: $\underline{b} \leq \underline{b}^*$ and $\overline{b}^* \leq \overline{b}$. By construction $\overline{a}$ and $\underline{a}$ are sharp bounds for $S(a^0)$.

3.3 Equilibrium analysis

One may wish to postulate that adaptive dynamics should result in equilibrium in the long run.[8] Under some seemingly natural conditions, the bounds $\underline{a}$ and $\overline{a}$ are in fact fixed points of F and thus our bounds are appropriate tools for equilibrium comparative statics. We start with a single-valued mapping f and then show how to extend the result to correspondence F.

Theorem 1. *Suppose $f : A \to A$ is an increasing mapping. If $f(\liminf_k b^k) = \liminf_k f(b^k)$ for every sequence $(b^k)_{k=0}^{\infty}$, then $\underline{a}$ is a fixed point. Similarly, if $f(\limsup_k b^k) = \limsup_k f(b^k)$ for every sequence $(b^k)_{k=0}^{\infty}$, then $\overline{a}$ is a fixed point.*

We will prove the first part of the theorem. The proof of the second part is analogous.

Proof. Note first that for any γ, we have $\bigwedge_{k \geq n-\gamma} \underline{a}^{k,\gamma} \leq \inf\{\underline{a}^{j-1,\gamma}, \underline{a}^{j-2,\gamma}, ..., \underline{a}^{j-\gamma,\gamma}\}$ for all $j \geq n \geq \gamma$. Indeed, all elements under the infimum on the right-hand side also appear under the infimum on the left-hand side. By an analogous argument, $\inf\{\underline{a}^{k-1,\gamma}, \underline{a}^{k-2,\gamma}, ..., \underline{a}^{k-\gamma,\gamma}\} \leq \underline{a}^{k-\gamma,\gamma}$. Letting $\underline{b}^{k,\gamma} := \inf\{\underline{a}^{k-1,\gamma}, \underline{a}^{k-2,\gamma}, \ldots, \underline{a}^{k-\gamma,\gamma}\}$,

$$\bigwedge_{k \geq n} \underline{b}^{k,\gamma} = \bigwedge_{k \geq n-\gamma} \underline{a}^{k,\gamma}. \tag{1}$$

for all $n \geq \gamma$. By the definition of $\underline{a}^{k,\gamma}$,

$$\liminf_{k \to \infty} \underline{a}^{k,\gamma} = \liminf_{k \to \infty} f(\underline{b}^{k,\gamma});$$

[8] We refer the reader to our related papers Balbus et al. (2025) and Olszewski (2021a,b) for results on tight fixed-point (equilibrium) bounds of best-response iterations on monotone correspondences and mappings.

by assumption,

$$\liminf_{k\to\infty} f(\underline{b}^{k,\gamma}) = f(\liminf_{k\to\infty} \underline{b}^{k,\gamma});$$

and by (1),

$$f(\liminf_{k\to\infty} \underline{b}^{k,\gamma}) = f(\liminf_{k\to\infty} \underline{a}^{k,\gamma}).$$

Therefore $\liminf_{k\to\infty} \underline{a}^{k,\gamma}$ is a fixed point, and so is $\underline{a}$ since f preserves lim inf's. □

Theorem 1 generalizes to weakly increasing correspondences $F : A \rightrightarrows A$ such that $\overline{F}(\limsup_k a^k) = \limsup_k \overline{F}(a^k)$ and $\underline{F}(\liminf_k a^k) = \liminf_k \underline{F}(a^k)$ by applying its present version for mappings to functions $f = \overline{F}$ and $f = \underline{F}$.

Continuity conditions in Theorem 1 imply downward and upward order continuity but are stronger. We conclude with an example that shows that $\underline{a}$ and $\overline{a}$ need not be fixed points in the general case, that is, for all order continuous mappings f.

Example 2. *Let $A = \{0,1\}^{\mathbb{N}}$ be endowed with the product order. For $a = (a_1, a_2, \ldots)$ define $f(a) := \big(\max(a_2, a_3), a_4, a_5, \ldots\big)$. In other words, $[f(a)]_1 = \max(a_2, a_3)$ and, for $t \geq 2$, $[f(a)]_t = a_{t+2}$. The set A is a complete lattice (with coordinatewise suprema and infima). Obviously, f is isotone and order continuous.*

Define $B := \big\{m! + \frac{1}{2}\big[(-1)^{m+1} + 1\big] : m \in \mathbb{N}\big\}$ and let $a_i^0 = 0$ for $i \in B$ and 1 otherwise. For $i \geq 2$ and $t \geq 1$ we claim that

$$a_i^{k,\gamma} = \prod_{r=\lceil t/\gamma \rceil}^{k} a_{i+2r}^0. \tag{2}$$

For $k = 1$, the recurrence gives $a_i^{1,\gamma} = a_{i+2}^0$, as required. We verify (2) inductively. Suppose that it holds for all positive $\tau \leq k-1$. From the definition of f and the induction hypothesis we get

$$a_i^{k,\gamma} = \prod_{j=1}^{\gamma} a_{i+2}^{k-j,\gamma} = \prod_{j=1}^{\gamma} \begin{cases} \displaystyle\prod_{r=\lceil (k-j)/\gamma \rceil}^{k-j} a_{i+2(r+1)}^0, & k-j \geq 1, \\ a_{i+2}^0, & k-j = 0. \end{cases} = \prod_{r=\lceil k/\gamma \rceil}^{k} a_{i+2r}^0.$$

In the last equality we used the fact that the indices from the preceding products, with repetitions omitted, fill exactly the integer interval $\{\lceil k/\gamma\rceil,\dots,k\}$. *Hence* (2) *is verified.*

Now, for each $i\geq 2$, *we construct arbitrarily large* k *such that* $a_i^{k,\gamma}=0$. *For sufficiently large* m *define*

$$t_m^i=\begin{cases}\dfrac{(2m)!-i}{2}, & i \text{ is even},\\[2mm] \dfrac{(2m+1)!+1-i}{2}, & i \text{ is odd}.\end{cases}$$

These are positive integers tending to infinity, and $i+2t_m^i\in B$. *Because the factor corresponding to* $r=t_m^i$ *occurs in* (2), *we obtain* $a_i^{t_m^i,\gamma}=0$ *for all sufficiently large* m. *Thus, for every* $i\geq 2$, *the equality* $a_i^{k,\gamma}=0$ *holds for infinitely many* k, *and consequently* $\liminf_k a_i^{k,\gamma}=0$ *for* $i\geq 2$.

Now we show that $a_1^{k,\gamma}=1$ *eventually in* k. *From the definition of* f *and* (2) *we have*

$$a_1^{k,\gamma}=\max\left(\prod_{r=\lfloor(k-1)/\gamma\rfloor}^{k-1}a_{2+2r}^0,\ \prod_{r=\lfloor(k-1)/\gamma\rfloor}^{k-1}a_{3+2r}^0\right). \tag{3}$$

Suppose, to the contrary, that $a_1^{k,\gamma}=0$ *for arbitrarily large* k. *Both products in* (3) *must then vanish. Write* $b_m:=m!+\frac{1}{2}\big[(-1)^{m+1}+1\big]$ *and put* $L_k:=\lfloor(k-1)/\gamma\rfloor$. *Consequently, there exist* $b_j,b_l\in B$ *such that* $2+2L_k\leq b_j\leq 2t$ *and* $3+2L_k\leq b_l\leq 2t+1$, *where* b_j *is even and* b_l *is odd. In particular* $b_j\neq b_l$. *For all sufficiently large* k, *these elements lie beyond the initial exceptional terms of* $(b_m)_{m=1}^{\infty}$, *and*

$$\frac{\max\{b_j,b_l\}}{\min\{b_j,b_l\}}\leq\frac{2k+1}{2+2L_k}\longrightarrow\gamma\qquad(t\to\infty).$$

On the other hand, the sequence $(b_m)_{m=1}^{\infty}$ *is eventually strictly increasing and* $\frac{b_{m+1}}{b_m}\longrightarrow\infty$. *Thus, for every fixed* γ, *all sufficiently late distinct terms of* $(b_m)_{m=1}^{\infty}$ *have their larger-to-smaller ratio greater than* $\gamma+1$. *Since* $L_k\to\infty$, *both* b_j *and* b_l *must be sufficiently late terms for large* k. *At the same time,* $(2k+1)/(2+2L_k)<\gamma+1$ *for all sufficiently large* k. *This contradiction shows that there exists* T_γ *such*

that $a_1^{k,\gamma} = 1$ *for all* $k \geq T_\gamma$. *As a result* $\underline{a}_1 = 1$ *and* $\underline{a}_i = 0$ *for* $i \geq 2$. *But* $f(\underline{a}) = \mathbf{0} \neq \underline{a}$.

4 Iterative monotone comparative statics

Consider now a poset of parameters T and a correspondence $F : A \times T \rightrightarrows A$. Let $\underline{a}(t)$ and $\overline{a}(t)$ be the bounds of adaptive learning sequences constructed in Section 3.2 starting from a^0 and iterating on $F(\cdot, t)$ for a given $t \in T$.

Theorem 2. *Let* A *be a sigma-complete lattice and* T *be a poset. Endow* $A \times T$ *with the product order. Let* $F : A \times T \rightrightarrows A$ *be such that* $F(a, t)$ *has the greatest and the least elements for each* a, t *and suppose* F *is weakly increasing on* $A \times T$. *If* $t < t'$, *then:*

(i) $\underline{a}(t) \leq \underline{a}(t')$ *and* $\overline{a}(t) \leq \overline{a}(t')$;

(ii) if a^0 *is a fixed point of* $F(\cdot, t)$ *and for any* $a \in A$, $F(a, \cdot)$ *is strongly increasing, then* $a^0 \leq \underline{a}(t')$;

(iii) in point (ii), if $\underline{F}(\cdot, t')$ *is, in addition, upward order continuous, then* $\underline{a}(t')$ *is the least fixed point of* $F(\cdot, t')$ *on* $\{a \in A : a \geq a^0\}$.

Point (i) assures that the bounds of iterations starting from any initial a^0 are ordered with respect to t. Weak monotonicity of F suffices to assure this weak monotone comparative statics result. Point (ii) provides a strong monotone comparative statics if iterations start from a fixed point a^0. Finally, point (iii) shows that under additional order continuity of $\underline{F}(\cdot, t')$ the lower bound of iterations $\underline{a}(t')$ is in fact a fixed point of $F(\cdot, t')$. Recall that all of this is in the setting of a sigma-complete lattice.

Proof. We start by proving (i). By the monotonicity of F, $\inf F(a^0,t) \leq \inf F(a^0,t')$. So, $\underline{a}^{1,\gamma}(t)$ is no greater than $\underline{a}^{1,\gamma}(t')$. Hence $\inf\{a^0, \underline{a}^{1,\gamma}(t)\} \leq \inf\{a^0, \underline{a}^{1,\gamma}(t')\}$, and $\underline{a}^{2,\gamma}(t) \leq \underline{a}^{2,\gamma}(t')$. By induction, $\underline{a}^{k,\gamma}$ obtained by iterating a^0 on $F(\cdot,t)$ is no greater than $\underline{a}^{k,\gamma}$ obtained by iterating a^0 on $F(\cdot,t')$ for any k. Hence, $\liminf_k \underline{a}^{k,\gamma}(t) \leq \liminf_k \underline{a}^{k,\gamma}(t')$ and hence $\underline{a}(t) \leq \underline{a}(t')$. We have proven the theorem for $\underline{a}$. The proof for $\overline{a}$ is analogous.

Proof of (ii). Now suppose a^0 is a fixed point of $F(\cdot,t)$. We will show, for all γ, that $a^0 \leq \underline{a}^{1,\gamma}(t')$ and the sequence $(\underline{a}^{k,\gamma}(t'))_{k=0}^{\infty}$ is increasing. Indeed, $a^0 \leq \overline{F}(a^0,t) \leq \underline{F}(a^0,t') = \underline{a}^{1,\gamma}(t')$ by strong monotonicity. If $(\underline{a}^{k,\gamma}(t'))_{k=0}^{n}$ is increasing, then

$$\begin{aligned}\underline{a}^{n+1,\gamma}(t') &= \underline{F}(\inf\{\underline{a}^{n-\gamma+1,\gamma}(t'), ..., \underline{a}^{n,\gamma}(t')\}, t') \geq \\ &\geq \underline{F}(\inf\{\underline{a}^{n-\gamma,\gamma}(t'), ..., \underline{a}^{n-1,\gamma}(t')\}, t') = \underline{a}^{n,\gamma}(t'),\end{aligned}$$

so $(\underline{a}^{k,\gamma}(t'))_{k=0}^{n+1}$ is increasing. By induction, we obtain $(\underline{a}^{k,\gamma}(t'))_{k=0}^{\infty}$ is increasing. Hence

$$a^0 \leq \liminf_k \underline{a}^{k,\gamma}(t') = \lim_k \underline{a}^{k,\gamma}(t')$$

for all γ. So, $a^0 \leq \underline{a}(t')$.

Proof of (iii). Since for each γ, $a^0 \leq \underline{a}^{1,\gamma}(t') = \underline{F}(a^0,t')$ hence $\underline{a}^{k,\gamma}(t')$ is constant equal $\underline{F}(a^0,t')$ on $k = 1,2,\ldots,\gamma$. By induction we can easily show that

$$\underline{a}^{n\gamma,\gamma}(t') \leq \underline{a}^{n\gamma+1,\gamma}(t') = \underline{a}^{n\gamma+2,\gamma}(t') = \ldots = \underline{a}^{n\gamma+\gamma,\gamma}(t') = \underline{F}(\underline{a}^{n\gamma,\gamma}(t'), t'),$$

for each $n \in \mathbb{N}$. Thus, $(\underline{a}^{k,\gamma}(t'))_{k=0}^{\infty}$ is increasing and $\underline{a}^{(k+1)\gamma,\gamma}(t') = \underline{F}(\underline{a}^{k\gamma,\gamma}(t'), t')$. Since $(\underline{a}^{k,\gamma}(t'))_{k=0}^{\infty}$ increasing, its supremum and limit is, say, $\underline{a}^{\gamma}(t')$. It is also the limit of $(\underline{a}^{k\gamma,\gamma}(t'))_{k=0}^{\infty}$. Applying upward order continuity we have $\underline{a}^{\gamma}(t') = \underline{F}(\underline{a}^{\gamma}(t'), t')$. Hence $\underline{a}^{\gamma}(t')$ is a fixed point of $\underline{F}(\cdot,t')$. Moreover $\underline{a}^{\gamma}(t')$ is independent of γ and so equal to $\underline{a}^{1}(t')$.

Now we only need to show that $\underline{a}^{1}(t')$ is the least fixed point of $F(\cdot,t')$ on the set $\{a \in A : a \geq a^0\}$. Consider an arbitrary $b \in A$ such that $b \geq a_0$ and that

$b \in F(b, t')$. Then for every n we obtain that $\underline{a}^{n,1}(t') \leq b$. For $n = 0$ the left-hand side is equal to a^0. Suppose that $\underline{a}^{k,1} \leq b$ for every $k \leq n$. Then we obtain

$$\underline{a}^{n+1,1}(t') = \underline{F}(\underline{a}^{n,1}(t'), t') \leq \underline{F}(b, t') \leq b$$

since $b \in F(b, t')$. As a result $\underline{a}^{n+1,1}(t') \leq b$ for every n. Hence $\underline{a}^1(t') \leq b$, and so $\underline{a}(t') \leq b$. □

Focusing on GSC allows us to compare our results to the corresponding results in Echenique (2002). Echenique's Theorem 3, as he introduces it, provides information about the limits of adaptive behavior after an increase in a parameter. Under his assumption that $a^0 \leq \inf F(a^0, t)$ made throughout his Theorem 3, the smallest equilibrium which is the limit of a convergent adaptive learning sequence $(a^k)_{k=0}^{\infty}$ coincides with our lower bound $\underline{a}(t)$ (see his Theorem 3.1. and 3.3.). By departing from his assumption, i.e. allowing that the adaptive behavior starts from an action profile a^0 that is not ordered with its image, we obtain the comparative statics of bounds of adaptive learning sequences. In applications, it is particularly relevant when the initial action profile is not an equilibrium.[9] Finally, with respect to Echenique Theorem 3.2: under the assumption that a^0 and $F(a^0)$ are ordered, although he allows for arbitrary γ in the definition of $S(a^0)$, his bounds from Theorem 3.2 do not depend on γ. Without this assumption, the bounds on adaptive sequences may depend on γ, and our construction allows for this possibility.

Moreover, observe that Theorem 2 applies to lattices A that are only sigma-complete. This is a generalization with respect to Echenique but also comparative statics results of Veinott (1992) (Theorem 2 in Chapter 10)[10] that require A to be a complete lattice, correspondence F to be strong set order monotone and pertain to comparative statics of extremal fixed points only. Again, Theorem 2 is

[9] If a^0 is an equilibrium, inequality $a^0 \leq \inf F(a^0, t)$ is trivially satisfied.

[10] See also Topkis (1998) Theorem 2.5.2.

particularly useful in environments in which comparative statics of iterative bounds is more adequate than comparative statics of extremal equilibria (see examples in Section 6).

Observe that Theorem 2 points (i) and (ii) do not require any (order) continuity of a correspondence F. This is in a stark difference with respect to Echenique Theorem 3 but also monotone comparative statics results based upon the application of Tarski-Kantorovich fixed point theorem on sigma-complete posets (see, e.g., Balbus et al. (2022) Proposition A.2.). So relaxing the demand for a theory of equilibrium comparative statics to comparative statics of iterative bounds, allows us to obtain a new comparative statics result for discontinuous correspondences.

Finally, Theorem 2, point (iii) shows that our approach delivers similar monotone comparative statics to that from Sabarwal (2025a) who studies order nearest fixed points for monotone correspondences. See his Theorem 3.2(a). The difference between his and our result lies in assumptions: he considers correspondences on complete lattices without imposing order continuity while we consider correspondences on sigma-complete lattices and (only in point (iii)) we impose upper order continuity.[11] Moreover, when A is a complete lattice our methods can be seen as providing constructive counterparts of Sabarwal (2025a) result[12], as his order nearest comparative statics are existential in nature while our comparative statics are based on accumulation points of adaptive learning sequences.

[11] In fact, as we analyze bounds for *any* adaptive learning sequences, in points (ii) and (iii), we also impose strong monotonicity of F in t.

[12] Let $\Phi(t)$ denote the set of fixed points of $F(\cdot, t)$. The most relevant ordering introduced in Sabarwal (2025a) for doing fixed point comparative statics for correspondences is the $*-$complete order relation that formalizes how to compare two nonempty subsets of a poset. What it means in the setting of our Theorem 2 is if A is a complete lattice, (a) for every fixed point $a^*(t) \in \Phi(t)$, there exists an order nearest fixed point $a^*(t') \in \Phi(t')$ such that $a^*(t) \leq a^*(t')$, and (b) for every fixed point $a^*(t') \in \Phi(t')$, there exists an order nearest fixed point $a^*(t) \in \Phi(t)$ again with $a^*(t) \leq a^*(t')$. Theorem 2 (ii) applied to fixed point iterations provides a computable version of Sabarwals (a). An obvious order dual version of Theorem 2 (ii) provides a computable version of (b). Of course, if A is only sigma complete, Sabarwal's set comparisons cannot be applied.

5 Aggregate comparative statics

In this section, we provide results on monotone "aggregate" comparative statics that concern the comparative statics of some statistics or aggregates φ of fundamental variables. These results differ from most results in the existing literature, which typically concern fundamental variables themselves, like our Theorem 2. One such situation was described in the Example 1, where the statistic φ was the aggregate (team) output and F was the joint best-response mapping. We will return to that example later in this section.

5.1 Defining monotone statistics

In applications, we are often interested in comparing some statistic $\varphi(a) \in \mathbb{R}$, where a in the long-run outcome for the original and new setting. More precisely, we are interested in comparing $\varphi(a^k)$ for large enough values of k. In our motivating example, φ was the output. Intuitively, if $\varphi(a^{new,k})$ in the new setting is at worst only slightly smaller than $\varphi(a^{old,k})$ for any large value of k in the original setting for any sequences $(a^{old,k})_{k=0}^{\infty}, (a^{new,k})_{k=0}^{\infty} \in \mathcal{S}(a^0)$, and this "slightly" is closer and closer to zero for larger k, then we can say that in the long-run the statistic φ weakly increases in response to the parameter change. In the realm of monotone mappings f or correspondences F, which we now restrict attention to, it makes sense to limit attention also to monotone statistics φ.

We are now ready to define monotone comparative statics in the most general case. Consider a correspondence $F : A \times T \rightrightarrows A$ and a statistic or an aggregate $\varphi : A \times T \to \mathbb{R}$. We now study only correspondences F and statistics φ that are monotone in a for any given t.

Definition 1. *A statistic $\varphi : A \times T \to \mathbb{R}$ weakly increases with a parameter change from t to t' if $\varphi(\underline{a}(t'), t') \geq \varphi(\underline{a}(t), t)$ as well as $\varphi(\overline{a}(t'), t') \geq \varphi(\overline{a}(t), t)$. We say such*

a statistic increases strongly with a parameter change if $\varphi(\underline{a}(t'),t') \geq \varphi(\overline{a}(t),t)$.

We can define in a dual manner when a statistic decreases. We will now motivate Definition 1. Suppose that we initially observe a^0 given a parameter t or t'. Let for a moment $\underline{a} := \underline{a}(t)$ and $\overline{a} := \overline{a}(t)$ and similarly $\underline{a}^{k,\gamma} := \underline{a}^{k,\gamma}(t)$ and $\overline{a}^{k,\gamma} := \overline{a}^{k,\gamma}(t)$. Recall that $\underline{a} \leq \liminf_k \underline{a}^{k,\gamma}$ for any γ, and that $\liminf_k \underline{a}^{k,\gamma}$ is the limit of the increasing sequence $\left(\inf_{l\geq k} \underline{a}^{l,\gamma}\right)_{l=0}^{\infty}$. Therefore, as we will consider order continuous comparative statics, for any $\varepsilon > 0$ and for any large enough value of k, it must be that $\varphi(\underline{a}) - \varepsilon < \varphi\left(\inf_{l\geq k} \underline{a}^{l,\gamma}\right)$, which in turn implies that $\varphi(\underline{a}) - \varepsilon < \varphi(\underline{a}^{l,\gamma})$ for $l \geq k$. Similarly, for any $\varepsilon > 0$ and for any large enough value of k, it must be that $\varphi(\overline{a}^{l,\gamma}) < \varphi(\overline{a}) + \varepsilon$ for $l \geq k$. In particular, $\varphi(\underline{a}) - \varepsilon < \varphi(a^k) < \varphi(\overline{a}) + \varepsilon$ for remote elements of any adaptive sequence $(a^k)_{k=0}^{\infty}$.

Therefore, if, for example, $\varphi(\underline{a}(t'),t') \geq \varphi(\overline{a}(t),t)$, then $\varphi(\overline{a}^{k,\gamma}(t),t)$ cannot exceed $\varphi(\underline{a}^{k,\gamma}(t'),t')$ by more than ε for any large value of k. So, "at infinity," $\varphi(a^k(t),t) \leq \varphi(a^k(t'),t')$ for any adaptive learning sequences $(a^k(t))_{k=0}^{\infty}$ and $(a^k(t'))_{k=0}^{\infty}$. Similarly, if $\varphi(\overline{a}(t'),t') \leq \varphi(\underline{a}(t),t)$, then $\varphi(a^k(t'),t')) \leq \varphi(a^k(t),t)$ at infinity for any adaptive sequences $(a^k(t'))_{k=0}^{\infty}$ and $(a^k(t))_{k=0}^{\infty}$. Similar in nature, but weaker in terms of results, interpretation can be provided for weak increase with a parameter change.

Note that our approach to comparative statics is conservative. Indeed, $b \leq \varphi(\underline{a})$ implies that $b - \varepsilon < \varphi(a^{k,\gamma})$ for large enough values of k. However, the converse is in general false. This converse implication would be true if we were comparing the elements of the lattice A instead a statistic thereof. More precisely, suppose that $b < \underline{a}^{k,\gamma}$ for large enough values of k. Then $b \leq \inf_{k\geq K} \underline{a}^{k,\gamma}$ for large values of k. So $b \leq \liminf_k \underline{a}^{k,\gamma}$ for all γ, and this implies that $b \leq \underline{a}$**.**

In the settings that satisfy the continuity conditions from Theorem 1, equilibrium comparative statics can be performed in the following way:

Definition 2. *Suppose a^0 is an equilibrium given some $t \in T$. A statistic $\varphi : A \times T \to \mathbb{R}$ increases in equilibrium with a parameter change from t to t' if $\varphi(\underline{a}, t') \geq \varphi(a^0, t)$, and it decreases with a parameter change if $\varphi(\overline{a}, t') \leq \varphi(a^0, t)$, where $\underline{a} = \underline{a}(t')$ and $\overline{a} = \overline{a}(t')$.*

5.2 Mixed shocks and aggregate comparative statics

As before, assume that A is a sigma-complete lattice and T a poset, and endow $A \times T$ with the product ordering. We will assume throughout the section that an aggregate $\varphi : A \times T \to \mathbb{R}$ is order continuous on A and monotone on $A \times T$. The mapping or correspondence $F : A \times T \to A$ is monotone on A, but in contrast to the previous section it is not necessarily monotone on T. For given t and t', let $(\underline{a}^{k,\gamma}(t))_{k=0}^{\infty}$, and $(\underline{a}^{k,\gamma}(t'))_{k=0}^{\infty}$ be the sequences from Section 3, such that $a^{0,\gamma}(t) = a^{0,\gamma}(t') = a^0 \in A$; and further, $\underline{a}^{k+1,\gamma}(t) = \underline{F}(\inf\{\underline{a}^{k,\gamma}(t), ..., \underline{a}^{k-\gamma+1,\gamma}(t)\}, t)$ and $\underline{a}^{k+1,\gamma}(t') = \underline{F}(\inf\{\underline{a}^{k,\gamma}(t'), ..., \underline{a}^{k-\gamma+1,\gamma}(t')\}, t')$. We start with conditions for weakly increasing statistics as defined in Definition 1.

Theorem 3. *Let $t' > t$. Suppose $\varphi(\bigwedge_{k\geq n} \underline{a}^{k,\gamma}(t'), t') \geq \bigwedge_{k\geq n} \varphi(\underline{a}^{k,\gamma}(t'), t')$ for any (sufficiently large) n and γ, and the following condition is satisfied:*

$$\text{if } \varphi(a,t) \leq \varphi(b,t') \text{ and } \varphi(a',t) \leq \varphi(b',t') \text{ for some } a, b, a', b' \text{ then}$$

$$\varphi\left(\underline{F}(a \wedge a', t), t\right) \leq \varphi\left(\underline{F}(b \wedge b', t'), t'\right). \tag{4}$$

Then we have $\varphi(\underline{a}(t), t) \leq \varphi(\underline{a}(t'), t')$.

Note that the hypotheses of Theorem 3, as well as the hypotheses of Theorem 4 below, concern only specific values of t and t' as well as specific elements of the adaptive sequences $(\underline{a}^k(t))_{k=0}^{\infty}$ and $(\underline{a}^k(t'))_{k=0}^{\infty}$. Condition $\varphi(\bigwedge_{k\geq n} \underline{a}^k(t'), t') \geq \bigwedge_{k\geq n} \varphi(\underline{a}^k(t'), t')$ is strong. However, as the example in Remark 1 below the proof shows, the theorem need not hold true when this condition is violated, even though

condition (4) is satisfied. Moreover, verifying this condition requires computing the sequence of iterations $(\underline{a}^k(t'))_{k=0}^{\infty}$. We stated the condition to pin down what we need for the proof. However, for many statistics φ even the stronger condition that $\varphi(\bigwedge B, t) = \bigwedge\{\varphi(b,t) : b \in B\}$ (for any $t \in T$ and any countable $B \subset A$) is satisfied, and this stronger condition does not require computing any sequence. A similar comment applies to Theorem 4. The stronger condition is satisfied in our leading example.

Theorem 3 (or Theorem 4) applies to the motivating example or, for example, when φ is a median and the vectors of an Euclidean space are ordered by comparing the smallest coordinate with the smallest coordinate, etc. In addition, our theorems seem to pin down the conditions needed for monotone comparative statics when F is not monotone on T.

Finally, it at first seems surprising that the condition (4) refers to only two pairs (a,b) and (a',b'), because γ can be any positive integer in the definition of general adaptive learning sequences. So, one would expect the present condition to refer to any finite set of pairs. It is possible to require less due to the important property of general adaptive learning sequences in Lemma 1 below. Before turning to more technical details, however, it might be helpful to illustrate Theorems 3 in the setting from our motivating example.

Motivating example, continued Recall that in the motivating example the best-reply mapping was $f(a_1,a_2,t) = (\frac{a_2}{t}, \min\{ta_1, 1\})$, and the aggregate was $\varphi(a_1,a_2,t) = 2\min\{ta_1, a_2\}$. Thus $\varphi(f(a,t),t) = 2\min\{a_2, \min\{ta_1,1\}\} = \varphi(a,t)$. We also have that $\varphi(\bigwedge B, t) = \bigwedge\{\varphi(b,t) : b \in B\}$ for any $B \subset A$, because both sides of the equality are equal to the output when each player provides the lowest effort across all her efforts in B. We will show that condition (4) from Theorem 3 is satisfied.

First, observe that:

$$\varphi(f(a \wedge a', t'), t') = \varphi(a \wedge a', t') =$$

$$= \varphi((\min\{a_1, a_1'\}, \min\{a_2, a_2'\}), t') = 2\min\{t'a_1, t'a_1', a_2, a_2'\}.$$

Now, $\varphi(a,t) \leq \varphi(b,t')$ and $\varphi(a',t) \leq \varphi(b',t')$, that is $\min\{ta_1, a_2\} \leq \min\{t'b_1, b_2\}$ and $\min\{ta_1', a_2'\} \leq \min\{t'b_1', b_2'\}$ imply that $\min\{ta_1, ta_1', a_2, a_2'\} \leq \min\{t'b_1, t'b_1', b_2, b_2'\}$, which means that $\varphi(f(a \wedge a', t), t) \leq \varphi(f(b \wedge b', t'), t')$. As a result Theorem 3 can be applied to obtain the iterative comparative statics result from the motivating example.

In case of our motivating example the result can also be verified directly. If $a_2^0/t > a_1^0$ (which is equivalent to $a_2^0 > ta_1^0$), then $\underline{a}_1^{1,\gamma} = \overline{a}_1^{1,\gamma} = a_1^1 = a_2^0/t$ and $\underline{a}_2^{1,\gamma} = \overline{a}_2^{1,\gamma} = a_2^1 = ta_1^0$. Therefore $(\underline{a}_1^{2,\gamma}, \underline{a}_2^{2,\gamma}) = (a_1^0, ta_1^0)$ for $\gamma \geq 2$, and $(\overline{a}_1^{2,\gamma}, \overline{a}_2^{2,\gamma}) = (a_2^0/t, a_2^0)$. Finally, $(\underline{a}_1^{k,\gamma}, \underline{a}_2^{k,\gamma}) = (\underline{a}_1^{2,\gamma}, \underline{a}_2^{2,\gamma})$ and $(\overline{a}_1^{k,\gamma}, \overline{a}_2^{k,\gamma}) = (\overline{a}_1^{2,\gamma}, \overline{a}_2^{2,\gamma})$ for $k > 2$ and all $\gamma \geq 2$.

If $a_1^0 > a_2^0/t$ (which is equivalent to $ta_1^0 > a_2^0$), then again $\underline{a}_1^{1,\gamma} = \overline{a}_1^{1,\gamma} = a_1^1 = a_2^0/t$ and $\underline{a}_2^{1,\gamma} = \overline{a}_2^{1,\gamma} = a_2^1 = \min\{ta_1^0, 1\}$. Therefore $(\underline{a}_1^{2,\gamma}, \underline{a}_2^{2,\gamma}) = (a_2^0/t, a_2^0)$ and $(\overline{a}_1^{2,\gamma}, \overline{a}_2^{2,\gamma}) = (\min\{a_1^0, 1/t\}, \min\{ta_1^0, 1\})$ for $\gamma \geq 2$. And again, $(\underline{a}_1^{k,\gamma}, \underline{a}_2^{k,\gamma}) = (\underline{a}_1^{2,\gamma}, \underline{a}_2^{2,\gamma})$ and $(\overline{a}_1^{k,\gamma}, \overline{a}_2^{k,\gamma}) = (\overline{a}_1^{2,\gamma}, \overline{a}_2^{2,\gamma})$ for $k > 2$ and all for $\gamma \geq 2$.

This gives us the following results: (a) If $a_2^0 \leq ta_1^0$, then the output in $\underline{a}(t > 1)$ is $2a_2^0$, which is equal to the total output in $\underline{a}(t = 1)$; (b) If $a_2^0 > ta_1^0$, then the output in $\underline{a}(t > 1)$ is $2ta_1^0$, which is greater than $2a_1^0$, the total output in $\underline{a}(t = 1)$. □.

As mentioned, in the proof of Theorem 3, an important role is played by the following lemma:

Lemma 1. *Let $\gamma \geq 2$ and consider $(\underline{a}^{k,\gamma})_{k=0}^{\infty}$, where $\underline{a}^{k+1,\gamma} = f(\bigwedge\{\underline{a}^{k,\gamma}, ..., \underline{a}^{k-\gamma+1,\gamma}\})$ for some increasing mapping $f : A \to A$ and given $a^{0,\gamma} \in A$. Then:*

1. *for any $n \geq 0$, $j \in \{1, ..., \gamma - 1\}$ we have $\underline{a}^{n\gamma+j,\gamma} = f(\underline{a}^{n\gamma,\gamma} \wedge \underline{a}^{n\gamma+j-1,\gamma})$,*

2. for any $n \geq 1$ we have $\underline{a}^{n\gamma,\gamma} = f(\underline{a}^{(n-1)\gamma,\gamma} \wedge \underline{a}^{n\gamma-1,\gamma})$.

Due to Lemma 1, we could state condition 4 in terms of pairs elements of a lattice (a and b, and a' and b') instead of any finite sequences of elements. At first sight, Lemma 1 is surprising, but as the reader can see in the proof, the two elements: $\underline{a}^{n\gamma,\gamma}$ and $\underline{a}^{n\gamma+j-1,\gamma}$ (or $\underline{a}^{(n-1)\gamma,\gamma}$ and $\underline{a}^{n\gamma-1,\gamma}$) encode everything contained in $\underline{a}^{k,\gamma}$ for $k < n\gamma + j$ that is necessary for computing $\underline{a}^{n\gamma+j,\gamma}$. We relegate the proof of Lemma 1 to Appendix, and we will proceed to proving Theorem 3.

Proof of Theorem 3. Since φ is monotone on T we have: $\varphi(a^0, t) \leq \varphi(a^0, t')$. By condition (4), taking $a = a' = a^{0,\gamma}(t)$ and $b = b' = a^{0,\gamma}(t')$ we obtain: $\varphi(\underline{a}^{1,\gamma}(t), t) \leq \varphi(\underline{a}^{1,\gamma}(t'), t')$. Applying condition (4) again we obtain: $\varphi(\underline{a}^{2,\gamma}(t), t) \leq \varphi(\underline{a}^{2,\gamma}(t'), t')$. Continuing by induction, if $\varphi(\underline{a}^{k-j,\gamma}(t), t) \leq \varphi(\underline{a}^{k-j,\gamma}(t'), t')$ for some k and each $j \in \{0, 1, \ldots, k\}$, then $\varphi(\underline{a}^{k+1,\gamma}(t), t) \leq \varphi(\underline{a}^{k+1,\gamma}(t'), t')$. Indeed, this follows from condition (4) and Lemma 1. As a result, we obtain: $\varphi(\underline{a}^{k,\gamma}(t), t) \leq \varphi(\underline{a}^{k,\gamma}(t'), t')$ for all k. Since $\bigwedge_{k\geq n} \underline{a}^{k,\gamma} \leq \underline{a}^{k,\gamma}$ for any $k \geq n$, by monotonicity of φ on A we have: $\varphi(\bigwedge_{k\geq n} \underline{a}^{k,\gamma}(t), t) \leq \bigwedge_{k\geq n} \varphi(\underline{a}^{k,\gamma}(t), t) \leq \bigwedge_{k\geq n} \varphi(\underline{a}^{k,\gamma}(t'), t') \leq \varphi(\bigwedge_{k\geq n} \underline{a}^{k,\gamma}(t'), t')$, where the last inequality follows from assumption: $\varphi(\bigwedge_k \underline{a}^{k,\gamma}(t'), t') \geq \bigwedge_k \varphi(\underline{a}^{k,\gamma}(t'), t')$. Taking the limit with n and γ and using the order continuity of φ on A we conclude that $\varphi(\underline{a}(t), t) \leq \varphi(\underline{a}(t'), t')$.□

Note that we used the monotonicity of φ on T only to obtain that $\varphi(a^{0,\gamma}(t), t) \leq \varphi(a^{0,\gamma}(t'), t')$.

Remark 1. *The following always holds: $\varphi(\bigwedge_{k\geq n} \underline{a}^{k,\gamma}(t'), t') \leq \bigwedge_{k\geq n} \varphi(\underline{a}^{k,\gamma}(t'), t')$, and so condition $\varphi(\bigwedge_{k\geq n} \underline{a}^{k,\gamma}(t'), t') \geq \bigwedge_{k\geq n} \varphi(\underline{a}^{k,\gamma}(t'), t')$ implies $\varphi(\bigwedge_{k\geq n} \underline{a}^{k,\gamma}(t'), t') = \bigwedge_{k\geq n} \varphi(\underline{a}^{k,\gamma}(t'), t')$. To see that this condition is critical, let us reconsider the sequence of best-response iterations (i.e. adaptive learning sequences with $\gamma = 1$) in our motivating example. Specifically, consider two sequences: $(a^k(t))_{k=0}^{\infty}$ and $(a^k(t'))_{k=0}^{\infty}$, where $a^0(t) = a^0(t') = (x, x)$ for some number $x \in (0, 1)$. When-*

ever $t' > 1$ *and* $t > 1$, *then* $a^k(t) = (\frac{x}{t}, \min\{xt, 1\})$ *for any odd* k *and* $a^k(t) = (\min\{x, \frac{1}{t}\}, x)$ *for any even* $k > 0$. *Similarly,* $a^k(t') = (\frac{x}{t'}, \min\{xt', 1\})$ *for any odd* k *and* $a^k(t') = (\min\{x, \frac{1}{t'}\}, x)$ *for any even* $k > 0$. *This implies that* $\bigwedge_{k\geq n} a^k(t) = (\frac{x}{t}, x)$ *and* $\bigwedge_{k\geq n} a^k(t') = (\frac{x}{t'}, x)$.

But let now the aggregate statistics be given by: $\varphi(a_1, a_2, t) = a_1 + a_2$. *Assume* $t' > t > 1$. *Then we have:* $\varphi(\bigwedge_{k\geq n} a^k(t'), t') = (1 + \frac{1}{t'})x < x + \min\{x, \frac{1}{t'}\} = \bigwedge_{k\geq n} \varphi(a^k(t'), t')$, *and so the assumption of Theorem 3 is violated. Observe that in this case the conclusion does not hold either. Indeed:* $\varphi(\underline{a}(t), t) = (1 + \frac{1}{t})x > (1 + \frac{1}{t'})x = \varphi(\underline{a}(t'), t')$,.

Remark 2. *Condition* $\varphi(\bigwedge_{k\geq n} \underline{a}^{k,\gamma}(t'), t') = \bigwedge_{k\geq n} \varphi(\underline{a}^{k,\gamma}(t'), t')$ *requires preservation of infima of sequences that are non-necessarily monotone. We now provide a general characterization of functions satisfying this condition. To simplify, we now drop parameters* t *and consider a function* $\varphi : A \to \mathbb{R}$, *where* $A = \prod_{i=1}^{n} [0, d_i]$ *with* d_i *real valued numbers and* A *endowed with a product order. We then have the following: for every increasing function* $\varphi : A \mapsto \mathbb{R}$ *satisfying*

$$\varphi(a \wedge a') = \min(\varphi(a), \varphi(a')) \tag{5}$$

for every $a, a' \in A$ *we have the following formula:*

$$\varphi(a) = \min(\eta_1(a_1), \eta_2(a_2), \ldots, \eta_n(a_n)), \tag{6}$$

where each $\eta_i : [0, d_i] \mapsto \mathbb{R}$ *is an increasing function. Clearly, also every function defined in (6) obeys (5). The condition (6) is therefore a characterization of (5) on* A.

To see that consider the following notation. For $a = (a_1, \ldots, a_n)$ *and* $a' = (a'_1, \ldots, a'_n)$ *and for* $I \subset \{1, 2, \ldots, n\}$ *define*

$$(a_{-I}, a'_I) = \begin{cases} a_i & if \quad i \notin I \\ a'_i & if \quad i \in I \end{cases}$$

In particular, denote $(a_{-i}, a'_i) = (a_{-\{i\}}, a'_{\{i\}})$ *and* $(a_{-i,-j}, a'_{i,j}) = (a_{-\{i,j\}}, a'_{\{i,j\}})$. *Define* $\eta_i(a_i) = \varphi(d_{-i}, a_i)$. *From (5) we have*

$$\varphi(d_{-i,-j}, a_i, a'_j) = \min\left(\varphi(d_{-i}, a_i), \varphi(d_{-j}, a'_j)\right) = \min(\eta_i(a_i), \eta_j(a'_j))$$

and suppose that for some $2 \leq k < n$ *and every* k *element set of indexes* $I_k := \{i_1, i_2, \ldots, i_k\}$ *we have*

$$\varphi(d_{-I_k}, a_{I_k}) = \min\left(\eta_{i_1}(a_{i_1}), \ldots, \eta_{i_k}(a_{i_k})\right). \tag{7}$$

We show the thesis for $I_{k+1} := \{i_1, i_2, \ldots, i_{k+1}\}$. *Let* $I_{2,k+1} := I_{k+1} \setminus \{i_1\}$. *Then from (5) and next from induction hypothesis in (7) we get*

$$\begin{aligned}
\varphi(d_{-I_{k+1}}, a_{I_{k+1}}) &= \min\left(\varphi(d_{-I_k}, a_{I_k}), \varphi(d_{-I_{2,k+1}}, a_{I_{2,k+1}}\right) \\
&= \min\left(\min\left(\eta_{i_1}(a_{i_1}), \ldots, \eta_{i_k}(a_{i_k})\right), \min\left(\eta_{i_2}(a_{i_2}), \ldots, \eta_{i_{k+1}}(a_{i_{k+1}})\right)\right) \\
&= \min(\eta_{i_1}(a_{i_1}), \ldots, \eta_{i_{k+1}}(a_{i_{k+1}})).
\end{aligned}$$

The thesis holds, hence, regardless of length of I_k *hence also for* $I_n := \{1, 2, \ldots, n\}$. *Hence (6) holds. In case of* $A = \prod_{i=1}^{\infty} [0, d_i]$ *condition (6) generalizes to* $\varphi(a) = \lim_{n\to\infty} \varphi(a_1, a_2, \ldots, a_n, d_{n+1}, d_{n+2}, \ldots)$ *if additionally* φ *is downward order continuous. Indeed, by the result for finite set we have* $\eta_i(a_i) = \varphi(d_{-i}, a_i)$.

Finally, if (5) holds for every pair a *and* a' *and* φ *is downward order continuous then* $\varphi(\bigwedge_{k\geq n} \underline{a}^{k,\gamma}) = \bigwedge_{k\geq n} \varphi(\underline{a}^{k,\gamma})$. *This gives an easy to check condition for the "continuity" type assumption required in Theorems 3 and 4. This condition seems natural for lower-envelope statistics such as Leontief output, bottleneck production, weakest-link settings, or Rawlsian welfare. See Section 6.2 for an additional application.*

A result analogous to that of Theorem 3 holds for upper bounds if we replace $\varphi(\bigwedge_{k\geq n} \underline{a}^{k,\gamma}(t'), t') \geq \bigwedge_{k\geq n} \varphi(\underline{a}^{k,\gamma}(t'), t')$ with $\varphi(\bigvee_{k\geq n} \overline{a}^{k,\gamma}(t'), t') \leq \bigvee_{k\geq n} \varphi(\overline{a}^{k,\gamma}(t'), t')$ and condition (4) with

$$\text{if } \varphi(a, t) \leq \varphi(b, t') \text{ and } \varphi(a', t) \leq \varphi(b', t') \text{ for some } a, b, a', b'$$

$$\varphi\left(\overline{F}(a \vee a', t), t\right) \leq \varphi\left(\overline{F}(b \vee b', t'), t'\right).$$

We then obtain: $\varphi(\overline{a}(t), t) \leq \varphi(\overline{a}(t'), t')$. For this analogous result and for Theorem 4, we need an analogue of Lemma 1 but for $(\overline{a}^{k,\gamma}(t'))_{k=0}^{\infty}$. We state it formally in the Appendix.

Our next and last theorem concerns the iterations started from a^0, a fixed point of $F(\cdot, t)$. It provides conditions for increasing aggregates according to Definition 2.

Theorem 4. *Let $t < t'$ and suppose $a^0 \in F(a^0, t)$; moreover $\varphi(\bigwedge_{k\geq n} \underline{a}^{k,\gamma}(t'), t') \geq \bigwedge_{k\geq n} \varphi(\underline{a}^{k,\gamma}(t'), t')$ for any (sufficiently large) n and γ, and the following condition is satisfied:*

$$\text{if } \varphi(a, t) \leq \varphi(b, t') \text{ and } \varphi(a', t) \leq \varphi(b', t') \text{ for some } a, b, a', b' \text{ then}$$

$$\varphi\left(\overline{F}(a \vee a', t), t\right) \leq \varphi\left(\underline{F}(b \wedge b', t'), t'\right). \tag{8}$$

Then we have $\varphi(a^0, t) \leq \varphi(\underline{a}(t'), t')$.

Proof of Theorem 4. From monotonicity of φ on T we have: $\varphi(a^0, t) \leq \varphi(a^0, t')$. By condition (8), taking $a = a' = a^0$ and $b = b' = a^0$, we obtain: $\varphi(\overline{a}^{1,\gamma}(t), t) \leq \varphi(\underline{a}^{1,\gamma}(t'), t')$. Applying condition (8) again, this time to $a = a^0$, $a' = \overline{a}^{1,\gamma}(t)$, and $b = a^0$, $b' = \underline{a}^{1,\gamma}(t')$, we obtain: $\varphi(\overline{a}^{2,\gamma}(t), t) \leq \varphi(\underline{a}^{2,\gamma}(t'), t')$. Continuing by induction, if $\varphi(\overline{a}^{k-j,\gamma}(t), t) \leq \varphi(\underline{a}^{k-j,\gamma}(t'), t')$ for some k and each $j \in \{0, 1, \ldots, k\}$ then $\varphi(\overline{a}^{k+1,\gamma}(t), t) \leq \varphi(\underline{a}^{k+1,\gamma}(t'), t')$. To see why, observe that $\overline{a}^{k+1,\gamma}(t) = \overline{F}(\overline{a}^{k,\gamma}(t) \vee \overline{a}^{l,\gamma}(t), t)$ by Lemma 2. Similarly, $\underline{a}^{k+1,\gamma}(t') = \underline{F}(\underline{a}^{k,\gamma}(t') \wedge \underline{a}^{l,\gamma}(t'), t')$ by Lemma 1. Note that l is the same in the formula for $\overline{a}^{k+1,\gamma}(t)$ and $\underline{a}^{k+1,\gamma}(t')$. Applying condition (8), $\varphi(\overline{a}^{k+1,\gamma}(t), t) \leq \varphi(\underline{a}^{k+1,\gamma}(t'), t')$. Observe that $(\overline{a}^{k,\gamma})_{k=0}^{\infty}(t)$ is increasing for all γ. Indeed, since $\overline{a}^{0,\gamma}(t) \leq \overline{F}(a^0, t) = \overline{a}^{1,\gamma}(t)$ and $\overline{a}^{k+1,\gamma}(t) = \overline{F}(\overline{a}^{k,\gamma}(t), t)$, $(\overline{a}^{k,\gamma}(t))_{k=0}^{\infty}$ is increasing by the monotonicity of $\overline{F}$ on A. Thus, $\varphi(a^0, t) \leq \varphi(\overline{a}^{n,\gamma}(t), t) \leq \varphi(\underline{a}^{n,\gamma}(t'), t')$ for all n and γ. Consequently,

$$\varphi(a^0, t) \leq \bigwedge_{k\geq n} \varphi(\underline{a}^{k,\gamma}(t'), t') \leq \varphi(\bigwedge_{k\geq n} \underline{a}^{k,\gamma}(t'), t'). \tag{9}$$

where the last inequality follows from continuity assumption on φ. Thus, $\varphi(a^0, t) \leq \varphi(\underline{a}(t'), t')$. $\square$

Motivating example, continued Finally, observe that, although condition (4) in Theorem 4 is not satisfied for arbitrary a, a', b, b',[13] it is satisfied for the elements of the two sequences $(\overline{a}^{k,\gamma}(t))_{k=0}^{\infty}$ and $(\underline{a}^{k,\gamma}(t'))_{k=0}^{\infty}$ to which condition (8) is applied in the proof of Theorem 4. Indeed, let $t' \geq t$ and consider a fixed point $a^0 = (\frac{x}{t}, x)$ (for some $x \in (0, 1]$) with the aggregate output of $2x$. Since $f(x/t, x) = (x/t, x)$ the sequence $(\overline{a}^{k,\gamma}(t))_{k=0}^{\infty}$ is constant. The sequence $(\underline{a}^{k,\gamma}(t'))_{k=0}^{\infty}$ starting from a^0 is given by $\underline{a}^{1,\gamma}(t') = (\frac{x}{t'}, \min\{\frac{xt'}{t}, 1\})$ and $\underline{a}^{2,\gamma}(t') = (\frac{x}{t'}, x)$ which is a fixed point of $f(\cdot, t')$ for $t' > t$. The value of the aggregate is: $\varphi(a^0, t) = 2x = \varphi(a^0, t')$ and $\varphi(\overline{a}^{1,\gamma}(t), t) = 2x = \varphi(\underline{a}^{1,\gamma}(t'), t')$, and $\varphi(f(a^0 \vee \overline{a}^{1,\gamma}(t), t), t) = \varphi(\overline{a}^{2,\gamma}(t), t) = 2x = \varphi(\underline{a}^{2,\gamma}(t'), t') = \varphi(f(a^0 \wedge \underline{a}^{1,\gamma}(t'), t'), t')$. The conclusion of Theorem 4 is hence that $\varphi(a^0, t) = 2x \leq 2x = \varphi(\underline{a}(t'), t')$ as stated in the motivating example. It is straightforward to show that $\varphi(\overline{a}(t'), t') = 2\min\{x\frac{t'}{t}, 1\}$.

6 Applications

6.1 Social learning on networks

Our first application is to studying social learning on networks. DeGroot's model, in which agents take weighted averages of the opinions they observe, is a commonly applied approach to studying social learning. Obviously, this very specific type of learning cannot well describe many real-life situations of interest. Cerreia-Vioglio et al. (2023) recently suggested a more general model, in which an opinion aggregator is a mapping that satisfies certain axioms. In their model of an economy

[13] Taking $a = (\frac{1}{2}, \frac{1}{4}), b = (\frac{1}{2}, 1), a' = (0, \frac{1}{4}), b' = (\frac{1}{2}, 0)$ we have that $\varphi(a, 1) = \frac{1}{2} \leq 2 = \varphi(b, 2)$ and $\varphi(a', 1) = 0 \leq 0 = \varphi(b', 2)$ but $\varphi(f(a \vee a', 1), 1) = \frac{1}{2} > 0 = \varphi(f(b \wedge b', 2), 2)$.

of n agents, the opinion profile is represented by a vector $a \in A = [0,1]^n$, and learning is represented by an opinion aggregator $F : [0,1]^n \to [0,1]^n$ that is monotone with respect to coordinate-by-coordinate ordering on $A = [0,1]^n$.[14] We will illustrate our approach to comparative statics by applying it to the setting studied by Cerreia-Vioglio et al. (2023).

Our first result says that when agents assign higher weights to higher opinions, then all agents have higher opinions in the long-run. More specifically, suppose that S is the set of all possible weight vectors, that is, $S = \{s = (s_1, ..., s_n) \in [0,1]^n : s_1 + ... + s_n = 1\}$, where s_i, the i-th coordinate of any s, represents the weight assigned to the i-th highest opinion. The set $T = S^n$ of weighting profiles (different agents may have different weight vectors) is partially ordered by $\preccurlyeq$, the coordinate-by-coordinate first-order stochastic dominance. Now, we consider two opinion aggregators $F : A \times T \to A$, which are defined as follows: each agent orders opinions in the input vector from the lowest to the highest, and her output opinion is the weighted average or the weighted median of the input opinions, where the agent uses her own weights.

Proposition 1. *If $t' \preccurlyeq t''$, then: $\underline{a}(t') \leq \underline{a}(t'')$ and $\overline{a}(t') \leq \overline{a}(t'')$; and if a^0 is a fixed point of $F(\cdot, t')$, then $a^0 \leq \underline{a}(t'')$.*

That is, if the agents shift their weights towards higher opinions, then the opinion of each of them increases in the long-run in response to this shift.

Note that this result holds true even though the opinions need not converge in the long run. Therefore, Proposition 1 could not have been obtained by referring to previously existing methods for performing comparative statics. In addition, it follows from the proof that F can be defined in a variety of other ways, for example,

[14] In addition to monotonicity, they impose two other axioms: normalization ($F(k, ..., k) = (k, ..., k)$ for all $k \in [0,1]$) and translation invariance ($F(a_1+k, ..., a_n+k) = F(a_1, ..., a_n)+(k, ..., k)$ whenever it makes sense). They all are satisfied in our application.

one may use the correspondence F that takes values between the weighted average and the weighted median, more precisely, $F(a) = \{b \in A : F^1(a) \wedge F^2(a) \leq b \leq F^1(a) \vee F^2(a)\}$, where F^1 is the weighted mean and F^2 is the weighted median.

Proof. We will check the assumption of Theorem 2. A is a complete lattice and hence sigma-complete. T is a poset. F is a function. To see that it is weakly increasing on $A \times T$ suppose that $a' \leq a''$ and $t' \preccurlyeq t''$. Then it follows directly from the definition of first-order stochastic dominance and the definition of F that $F(a', t') \leq F(a', t'')$. To complete the proof we must show that $F(a', t'') \leq F(a'', t'')$. We will show that $(a')^{(k)} \leq (a'')^{(k)}$, where $a^{(k)}$ denotes the k-th lowest coordinate of a, and this will complete the proof. Indeed, if the $k-1$ lowest coordinates of a' are the same as the $k-1$ lowest coordinates of a'', then $(a')^{(k)}$ is the lowest of the remaining $n-k+1$ coordinates of a', and $(a'')^{(k)}$ is the lowest of the corresponding $n-k+1$ coordinates of a''. Then, $(a')^{(k)} \leq (a'')^{(k)}$, because $a' \leq a''$ in the coordinate-by-coordinate ordering. Otherwise, $(a'')^{(l)}$ for an $l < k$, is the coordinate of a'' corresponding to one of the $n-k+1$ highest coordinates of a'. This coordinate of a'' is higher than the coordinate of a' it corresponds to, and so $(a'')^{(l)} \geq (a')^{(k)}$. This completes the proof, because $(a'')^{(k)} \geq (a'')^{(l)}$ by definition. □

The above result cannot be obtained using previous results. To stress just one difference with respect to the literature, observe that our comparative statics is *local* as it concerns bounds of adaptive learning sequences starting from *any initial opinion profile*, neither extremal nor necessarily equilibrium one.

We finally illustrate by an example that our approach allows for comparative statics even *beyond* of what is covered in our results. More specifically, one may be interested how different initial opinions affect the opinions in the long-run. So, consider a group of agents $N = \{1, 2, 3\}$ who share their opinions $a^0 \in [0, 1]^3$.

Suppose that the weights assigned to the other agents are represented by the matrix

$$W = \begin{bmatrix} 0.4 & 0.3 & 0.3 \\ 0.1 & 0.3 & 0.6 \\ 0.1 & 0.6 & 0.3 \end{bmatrix}.$$

Note that this time agents assign weights by identity not according to the ranking of opinions, that is, the entry in column j and row i of the matrix represents the weight assigned by agent i to the opinion of agent j.

Consider the aggregation induced by the median. For example, $a^1 = F(a^0) = (0.6, 0.6, 0.6)$ for $a^0 = (0.6, 0.6, 0.6)$ and $a^1 = F(a^0) = (0.6, 0.4, 0.6)$ for $a^0 = (0.8, 0.6, 0.4)$. The median aggregator satisfies the conditions required by of Cerreia-Vioglio et al. Actually, the aggregator was used as an example in their paper.

When $a^0 = (0.6, 0.6, 0.6)$, then $\underline{a} = \overline{a} = (0.6, 0.6, 0.6)$. When $a^0 = (0.8, 0.6, 0.4)$, then $\underline{a}^{k,1} = \overline{a}^{k,1} = (0.6, 0.4, 0.6)$ for odd k and $\underline{a}^{k,1} = \overline{a}^{k,1} = (0.6, 0.6, 0.4)$ for even k. Thus, $\liminf_{k=\infty} \underline{a}^{k,1} = (0.6, 0.4, 0.4)$ and $\limsup_{k=\infty} \overline{a}^{k,1} = (0.6, 0.6, 0.6)$. For $\gamma \geq 2$, $\underline{a}^{1,\gamma} = \overline{a}^{1,\gamma} = (0.6, 0.4, 0.6)$; and for $k \geq 2$, $\underline{a}^{k,\gamma} = (0.4, 0.4, 0.4)$ and $\overline{a}^{k,\gamma} = (0.6, 0.6, 0.6)$. So, $\liminf_{k=\infty} \underline{a}^{k,\gamma} = (0.4, 0.4, 0.4)$ and $\limsup_{k=\infty} \overline{a}^{k,\gamma} = (0.6, 0.6, 0.6)$. This gives $\underline{a} = (0.4, 0.4, 0.4)$ and $\overline{a} = (0.6, 0.6, 0.6)$.

Therefore we conclude that if the initial opinions change from $(0.6, 0.6, 0.6)$ to $(0.8, 0.6, 0.4)$, then the opinions of all agents cannot increase in the long run. Intuitively, the reason is that the opinions of agent 1 are less influential than the opinions of agent 3.

6.2 Supply chain networks

Let $I = [0, 1]$ be an interval of firms distributed according to a Lebesgue measure λ. Firms are involved in a supply chain network. Each firm (supplier) i chooses

$\hat{a}_i \in K$ (for some compact $K \subset \mathbb{R}$) denoting a capacity / investment in own reliability (or quality). Let a be a measurable profile of strategies $a : I \mapsto K$. We restrict attention of Lipschitz continuous profiles denoted by A.

The effective capacity of the supply chain to firm i is determined by the reliability of its weakest supplier. The overall payoff of each firm is hence

$$u_i(\hat{a}_i, a, t) := -0.5 \left(\hat{a}_i - \alpha - t\sigma(i) - \gamma \left(\rho t + \operatorname{ess\,inf}_{i \in I} a(i) \right) \right)^2,$$

where α, γ, t are some real valued parameters and $\sigma : I \mapsto \mathbb{R}$ is increasing and Lipschitz continuous function. Assume $\gamma \in (0, 1)$ and $\rho > 0$. The game is of strategic complementarities: investments by other suppliers increase the return to investments by other suppliers, e.g. because compatible quality standards or capacity coordination are needed.

Suppose K is large enough so that argmax for each firm is interior. Compute best response of each firm and denote it by:

$$f(a, t)(i) := \alpha + t\sigma(i) + \gamma(\rho t + \operatorname{ess\,inf}_{j \in I} a(j)).$$

Assume that $\operatorname{ess\,inf}_{i \in I} \sigma(i) + \rho\gamma < 0 < \operatorname{ess\,sup}_{i \in I} \sigma(i) + \rho\gamma$, that is shock t has different effect on investment incentives of each suppliers, e.g. it increases investment incentives of some (these with positive $\sigma(i)$) but decreases incentives of other suppliers (these with sufficiently negative $\sigma(i)$.

We are interested in the comparative statics of the aggregate capacity / reliability given by: $\varphi(a, t) = \rho t + \operatorname{ess\,inf}_{i \in I} a(i)$ with respect to parameter $t \geq 0$, e.g. a shock that directly increases overall capacity / reliability. Let strategy $\underline{a}(t) : I \to \mathbb{R}$ denote the bound of adaptive learning for parameter t.

Proposition 2. *If* $\rho + ess\operatorname{inf}_{i \in I} \sigma(i) > 0$, *then* $\varphi(\underline{a}(t), t) \leq \varphi(\underline{a}(t'), t')$.

The above result is new with respect to the literature, and not covered by previous available results in the literature. In particular, it holds even though the

parameter (aggregate capacity / reliability) change is a "mixed shock" to the best response of individual firms.

Proof. The action space A is a set of measurable, essentially bounded, real valued functions ordered almost everywhere. It is a complete lattice. $T \subset \mathbb{R}$ is a poset. The aggregate $\varphi : A \times T \to \mathbb{R}$ is downward order continuous on A and monotone on $A \times T$. The best response mapping $f : A \times T \to A$ is monotone on A but it is not necessarily monotone on T. Moreover, since σ is Lipschitz continuous on I and $f(a,t)$ is increasing on I one can show that $\varphi(\bigvee_k f^k(a,t),t) = \bigvee_k \varphi(f^k(a,t),t)$ for any increasing sequence of functions $(f^k(a,t))_{k=0}^{\infty}$. We now verify conditions in Theorem 3.

First, observe that function $\varphi(\cdot,t)$ satisfies condition (6) from Remark 2, and hence $\varphi(\min\{a,a'\},t) = \min\{\varphi(a,t),\varphi(a',t)\}$. Second, compute

$$\varphi(f(a,t),t) = \rho t + \operatorname{ess}\inf_{i\in I} f(a,t)(i) = \alpha + \gamma\varphi(a,t) + t(\rho + \operatorname{ess}\inf_{i\in I}\sigma(i)),$$

Observe that the aggregate capacity / reliability evaluated at the best response depends on the firm with the weakest (and negative) investment incentives. This formula above allows us to directly verify condition (4) in Theorem 3, namely: if $\varphi(a,t) \le \varphi(b,t')$ and $\varphi(a',t) \le \varphi(b',t')$ then

$$\varphi(f(\min\{a,a'\},t),t) \le \varphi(f(\min\{b,b'\},t'),t').$$

Indeed:

$$\varphi(f(\min\{a,a'\},t),t) = \alpha + \gamma\varphi(\min\{a,a'\},t) + t(\rho + \operatorname{ess}\inf_{i\in[0,1]}\sigma(i))) =$$

$$\alpha + \gamma\min\{\varphi(a,t),\varphi(a',t)\} + t(\rho + \operatorname{ess}\inf_{i\in[0,1]}\sigma(i)).$$

Noting that, by the hypothesis $\min\{\varphi(a,t),\varphi(a',t)\} \le \min\{\varphi(b,t'),\varphi(b',t')\}$ we obtain the desired inequality whenever $\rho + \operatorname{ess}\inf_{i\in[0,1]}\sigma(i) > 0$.

□

Remark 3. *The above example illustrates a useful sufficient condition for implication (4) to hold. Assume* $\varphi(\min\{a, a'\}, t) = \min\{\varphi(a, t), \varphi(a', t)\}$ *as well as that*

$$\varphi(f(a, t), t) = H(\varphi(a, t), t)$$

for some increasing function $H : \mathbb{R} \times T \to \mathbb{R}$ *and for each* t *and* a, a'*. Then implication (4) holds. Indeed, suppose* $t' > t$ *and* $\varphi(a, t) \leq \varphi(b, t')$ *and* $\varphi(a', t) \leq \varphi(b', t')$*. By the first assertion we have* $\varphi(\min\{a, a'\}, t) = \min\{\varphi(a, t), \varphi(a', t)\}$ *and similarly* $\varphi(\min\{b, b'\}, t') = \min\{\varphi(b, t'), \varphi((b', t')\}$*. Hence by the supposition in implication (4) we get:*

$$\varphi(\min\{a, a'\}, t) \leq \varphi(\min\{b, b'\}, t').$$

Since H *is increasing in both arguments this means:* $H\left(\varphi(\min\{a, a'\}, t), t\right) \leq H\left(\varphi(\min\{b, b'\}, t'), t'\right)$*. Now using the second assertion:* $\varphi(f(a, t), t) = H(\varphi(a, t), t)$ *we get exactly the desired condition:*

$$\varphi\left(f(\min\{a, a'\}, t), t\right) \leq \varphi\left(f(\min\{b, b'\}, t'), t'\right).$$

6.3 Interim Bayesian supermodular games

Our approach provides new results on the existence and comparative statics of pure strategy equilibria in the interim formulation of Bayesian supermodular games.[15] The primitives of the game are given by a tuple $(I, T, (\mathcal{F}_i, A_i, u_i, p_i)_{i \in I}, \Theta)$, where I is the set of players. The types are given by T_i with sigma-field $\mathcal{F}_i$ for each $i \in I$ (where we let $T = \prod_{i \in \{0\} \cup I} T_i$ with the product sigma-field). The component T_0 (with sigma-field $\mathcal{F}_0$) represents residual uncertainty not observed by the players

[15] The class of interim Bayesian games is broader than the ex-ante Bayesian games. Moreover, the notion of interim Bayesian Nash equilibrium is stronger than the notion of an ex-ante Bayesian Nash equilibrium. See Van Zandt (2010) for details and the global games literature (e.g., Morris and Shin (2002) and Morris et al. (2016)) for applications.

(or the set of possible payoffs). Type dependent belief functions for the players are given by $p_i : T_i \to M_{-i}$, where M_{-i} is the set of probability measures on T_{-i} with the product sigma-field $\mathcal{F}_{-i}$. For each set $F_{-i} \in \mathcal{F}_{-i}$, the mapping $t_t \to p_i(F_{-i} \mid t_i)$ is measurable. The action set for each $i \in I$ is A_i, which is a compact metric lattice (with its Borel sigma-field), and we denote the space of joint action profiles by A. Finally, Θ is a partially ordered set of parameters.

The players' payoffs are given by $u_i : A \times T \times \Theta \to \mathbb{R}$ with u_i bounded. We additionally assume: (a) u_i is continuous and supermodular in player's own action, (b) u_i is measurable in t, (c) u_i has increasing differences between (a_i, a_{-i}), and (d) u_i has (i) increasing differences (resp, (ii) strictly increasing differences) between (a_i, θ). Equilibria of this game were studied by Van Zandt (2010).

The set of interim Bayesian Nash equilibrium (BNE) at each parameter $\theta \in \Theta$ coincides with the set of fixed-points of the correspondence defined by the players' joint best response correspondences. That is, the BNE are fixed points of a parameterized, weakly increasing correspondence $F : S \times \Theta \rightrightarrows S$, where S denotes the set of strategies - measurable functions mapping individual types to individual actions (a sigma-complete lattice) - endowed with the pointwise partial order.

Our Theorem 2, applied to the best response correspondence F, provides new results on equilibrium comparative statics with respect to θ.

Proposition 3. *Under assumptions (a)-(d)(i), for any initial profile of strategies the bounds of adaptive learning sequences iterated on a best response correspondence with higher parameter are higher than the bounds of adaptive learning sequences iterated on a best response correspondence with lower parameter. Moreover, under condition (d)(ii) (instead of (d)(i)) adaptive learning sequences started from an initial interim BNE for a smaller parameter, but iterated on a best response correspondence at a higher parameter, are bounded below by an order nearest interim BNE of a game with higher parameter.*

Relative to Van Zandt (2010), our results provide iterative monotone comparative statics starting from *any initial profile* of strategies, in particular, even when that profile is not an equilibrium profile. Moreover, since the space of measurable functions S is not a complete lattice, the results in Sabarwal (2025a) do not apply.

Proof. S is a sigma-complete lattice with greatest and least elements.[16] Indeed, for a countable sequence of measurable functions, their pointwise supremum and infimum are measurable. Next, under assumptions (a)-(d) the best response correspondence in weakly monotone. It also admits least and greatest selections that are, respectively, upward order continuous and downward order continuous (see Van Zandt (2010)). As a result Theorem 2 (i) guarantees the first assertion. The second follows from Theorem 2 (iii). □

6.4 Long-run income distribution in monotone economies

Our results can be applied to the comparative statics of long-run distributions of output or income associated with more abstract infinite-horizon stochastic growth models with nonconvexities. Let the production or income available at period t be $y_t \in Y$, where $Y = [0, \bar{Y}] \subset \mathbb{R}_+$. Agent selects a consumption level $c_t \in [0, y_t]$, with the remaining resources $i_t = z_t - c_t$ allocated as an investment. The evolution of income is given by $y_{t+1} = f(i_t, z_{t+1})$ where f is a continuous, strictly increasing production function and z_{t+1} is a random shock drawn each period from distribution π over a finite set Z. For simplicity, we assume full depreciation. The temporal utility is given by a continuous, strictly increasing and strictly concave

[16] In the ex-ante formulation of our Bayesian supermodular game (e.g., see van Zandt and Vives (2007), as well as Sabarwal (2025b)), one works with equivalence classes of measurable strategies ordered almost everywhere pointwise. The set of such strategies is a complete lattice. Existence of ex-ante pure strategy BNE then follows from an application of the Veinott-Zhou version of Tarski's fixed point theorem and order-nearest equilibrium comparative statics follow from recent results in Sabarwal (2025a).

mapping $u : Y \to \mathbb{R}$. The agent's objective then is to maximize her expected discounted payoffs over an infinite horizon, given an initial state $y_0 \in Y$ and discount $\beta \in (0,1)$. Denote the value of this optimization problem by $v^*_\beta(y_0)$. This problem admits a recursive representation, where $v = v^*_\beta$ is the unique solution to the Bellman equation: $v(z) = \max_{i \in [0,y]} u(y-i) + \beta \int_Z v(f(i,z'))d\pi(z')$.

Let the policy correspondence be given by $H^*(y,\beta) = \arg\max_{i \in [0,y]} u(y-i) + \beta \int_Z v^*_\beta(f(i,z'))d\pi(z')$. By $\mathcal{M}(Y)$ denote a set of measures on Y endowed with the first-order stochastic dominance and the weak convergence of measures. For a measurable set $B \subset Y$ define the stochastic income transition with $Q(B|i) := \int_Z 1_B(f(i,z'))d\pi(dz')$. For any selector $h_\beta(\cdot) \in H^*(\cdot,\beta)$, define the associated adjoint Markov operator: $\Lambda_{h_\beta}\mu(B) = \int_Y Q(B|h_\beta(y))\mu(dy)$ and the associated adjoint Markov correspondence: $\Lambda\mu(B) = \{\Lambda_{h_\beta}\mu(B)\}_{h_\beta \in H^*(\cdot,\beta)}$.

We can now apply our Theorem 2 to characterize the iterative monotone comparative statics of the stationary income distributions.

Proposition 4. *When $\beta_1 \leq \beta_2$, for any initial measure $\mu_0 \in \mathcal{M}(Y)$, the lower (resp., upper) bounds for long-run income distribution dynamics are increasing, i.e. $\underline{\mu}(\beta_1) \leq \underline{\mu}(\beta_2)$ (resp., $\overline{\mu}(\beta_1) \leq \overline{\mu}(\beta_2)$), and from any stationary equilibrium at the discount rate β_1, say μ_{β_1}, we conclude that iterations on Λ_{β_2} from μ_{β_1} satisfy $\mu_{\beta_1} \leq \underline{\mu}(\beta_2)$.*

Proof. First, $\mathcal{M}(Y)$ is a complete lattice (See, for example, see Kamae et al. (1977)). Second, by standard arguments[17] $(y,\beta) \to v^*_\beta(y)$ has increasing differences. Third, since u is strictly increasing and strictly concave, the objective in the Bellman equation has strictly increasing differences in $(i; y, \beta)$. Then, by an

[17] Observing that maximization in the Bellman equation preserves increasing differences and then applying the Banach fixed point theorem. See Hopenhayn and Prescott (1992) Lemma 1 or Balbus et al. (2022) Lemma A.1.

application of the Topkis (1998) Theorem (e.g., Theorem 2.8.4), the policy correspondence[18] H^* is a nonempty and jointly strongly increasing in (y, β). It also possess the greatest and the least selections. By Huggett (2003) Theorem 1, since H^* is strongly increasing, Λ is strongly increasing on $M(Y)$. As H^* has the extremal selections, so does Λ. Applying Theorem 2 we obtain the results. □

We remark that a similar reasoning can be applied to study the stationary equilibrium distribution in large dynamic economies in the spirit of Bewley or Huggett/Aiyagari models without aggregate risk. Indeed, interpreting μ as a distributions of income over Y in some large economy, we can study monotone comparative statics of stationary or invariant income distributions after the monotone exogenous shock to the policy function h in the income fluctuation problem of the shocks governed by Q for any initial income distribution μ^0.

In addition, our results extend the stationary equilibrium comparative statics for monotone economies based upon the work of Hopenhayn and Prescott (1992), Huggett (2003), and Acemoglu and Jensen (2015) as they impose either unique (and stable) stationary equilibrium distribution or, under multiplicity, concentrate on extremal stationary equilibrium distributions only, that could be trivial or far away from distributions relevant for economic analysis.

6.5 Comparing recursive equilibria in dynamic models with indeterminate equilibria

We finally show how to apply our results to obtain monotone comparative statics of (minimal state space) recursive equilibria (RE) in macroeconomic models with multiplicities.[19] Consider a simple stochastic OLG economy with production (e.g., as in McGovern et al. (2013)). There is a continuum of identical agents born each

[18] Recall, since f is not necessarily concave in i the policy is not necessarily unique.

[19] See Coleman (1991) and Mirman et al. (2008) for motivation and other references.

period who live for two periods. Let s and s' denote vectors of aggregate state variables in the current and the following periods (to be specified in a moment). In the first period of life, agents are endowed with a unit of time which they supply inelastically to the firm at the prevailing wage $w(s)$, and they consume and save. In the second period of life, agents consume their savings which are subjected to a stochastic return $r(s')$. Preferences are time separable with discount rate $\beta \in (0,1)$ and are given by $u(c_1) + \beta v(c_2)$, where consumption when young (resp., old) is denoted by c_1 (resp., c_2), and $u : \mathbb{R}_+ \to \mathbb{R}$ and $v : \mathbb{R}_+ \to \mathbb{R}$ are smooth, strictly increasing, strictly concave, with $\lim_{c\to 0^+} u'(c) = \infty = \lim_{c\to 0^+} v'(c)$.

The reduced-form technology is given by $f(k,n)e(K,N,z)$ where f is a technology transforming private inputs of capital and labor (k,n), and the externality $e(K,N,z)$ is a total factor productivity that depends on per capital aggregates of capital and labor (K,N) and a shock $z \in Z = [z_l, z_h] \subset \mathbb{R}_{++}$ which is drawn each period from a first-order Markov process with stationary transition $\pi(z,z')$ that satisfies a Feller property. We let f satisfy typical assumptions, namely: it is constant returns to scale, increasing (but increasing strictly with each argument for the positive input of the other), concave jointly (but strictly concave with each argument separately for the positive input of the other), twice continuously differentiable and satisfies an Inada condition $\lim_{k\to 0} f_1(k,1) \to \infty$. Moreover, $e(K,1,z)$ is such that $f_1(k,1)e(K,1,z)$ is decreasing in k and increasing in K for $K > 0$ such that $\lim_{k\to 0} f_1(k,1)e(k,1,z) = r(k,z) \to \infty$,[20] $\lim_{k\to 0^+} r(k, z_{\max})k = 0$; $w(k,k,z) := f_2(k,1)e(k,1,z)$ is increasing in k with $\lim_{k\to 0^+} w(k,z) = 0$; both r and w are increasing in z for all k. Finally, $f(0,1)e(0,1,z) = 0$ for any z and there

[20] This model is known to have multiple equilibria. Recall $f_1(k,1)e(K,1,z)$ under our assumptions is *mixed monotone* in (k,K), i.e., decreasing in k and increasing in K. This is a critical feature that creates the possibility for equilibrium indeterminacy (i.e. a continuum of equilibria) in this class of models (see Santos (2002) and especially Datta et al. (2018) for discussion). Also, note our arguments here work under the alternative assumption that $\lim_{k\to 0} f_1(k,1)e(K,1,z) \to \infty$ for all $K > 0$. See Datta et al. (2018).

exists a maximal sustainable capital stock[21] denoted by $k_{\max}$. This last assumptions ensure that the state space for the equilibrium capital stock lies a compact set $X \subset \mathbb{R}_+$. Many specific examples of technologies that satisfy these assumptions can be given (see, e.g., Datta et al. (2018)).

We now proceed to define a RE. Anticipating that $n = 1 = N$ and $k = K$ in any RE, and denoting the aggregate vector of state variables by $s = (K, z) \in S = X \times Z$, we consider the existence of RE in a class of investment functions W with pointwise partial orders. In particular, let $W = \{h : S \to \mathbb{R}_+, 0 \leq h \leq w, h(k, z)$ increasing in k, and measurable in $(k, z)\}$ where W is endowed with pointwise partial order. Together with π, $h \in W$ describes the law of motion for the aggregate variables. Taking this, a young agent solves: $\max_{y \in [0, w(s)]} u(w(s)-y)+\beta \int_Z v(r(h(s), z')y)\pi(z, dz')$. Let $\hat{y}(s; h)$ be the optimal solution to this household problem for $h \in W$. The optimal solution is unique under our assumptions. Labor and capital markets are competitive hence by profit maximization, in an RE, $w(K, z) = f_2(K, 1)e(K, 1, z)$ and $r(K, z) = f_1(K, 1)e(K, 1, z)$. An RE of this economy in the space W is a law of motion $h^* \in W$ and policy function $y^* \in W$ such that $y^*(s) = \hat{y}(s; h^*) = h^*(s) \in W$ for each $s \in S_{++}$ whenever $h^*(s) > 0$. Here $S_{++} := X_{++} \times Z$ with $X_{++} \subset \mathbb{R}_{++}$. Market clearing is implied by the formulation of the household problem.

We now proceed to conduct a comparative statics exercise. Specifically, consider the question of capital deepening in the discount rate β to the set of RE in this economy. Define a nonlinear operator F_β on W as follows: for $h(s) > 0$, $h \in W$, let the operator $F_\beta h(s)$ be the unique $y = y^*(s; h)$ solving:

$$Z_\beta(y; s, h) = u'(w(s) - y) - \beta \int_Z v'(r(h(y, z')), z')y)r(y, z')\pi(z, dz') = 0, \quad (10)$$

with $F_\beta h(s) = 0$, whenever $h(s) = 0$ in any state s. Any fixed point of F_β, say

[21] $\forall k \geq k_{\max}$ and $\forall z \in Z$, $F(k, 1, k, 1, z) \leq k_{\max}$.

$h^*_\beta \in W$, is an RE law of motion if and only if it is a non-zero fixed point of the operator F_β in W. We apply Theorem 2 to the operator F_β.

Proposition 5. *For any initial $h^0 \in A$ and $\beta_1 \leq \beta_2$, we have: (i) the lower bounds for RE satisfy $\underline{h}(\beta_1) \leq \underline{h}(\beta_2)$ and the upper bounds satisfy $\overline{h}(\beta_1) \leq \overline{h}(\beta_2)$, (ii) taking initially any RE $h^*_{\beta_1} \in A$ but iterating on F_{β_2} we also have the RE comparative statics $h^*_{\beta_1} \leq \underline{h}(\beta_2)$.*

Proof. First, W is a sigma-complete.[22] Notice for each $\beta \in (0,1)$, $h \in W$, $F_\beta h$ is well defined. To see this, when $h = 0$ in any state, $F_\beta h$ =0. When $h > 0$, then by Inada conditions on u, as $y \to w$, $Z_\beta(y, s, h) \to \infty$ (as the first term $\to \infty$, and the second term is finite), and as $y \to 0$, $Z(y; s, h) \to -\infty$ (as the second term $\to -\infty$, and the first term is finite). As Z is strictly increasing and continuous in y under our assumptions, $F_\beta h$ exists and is unique. Also, for each $\beta \in (0,1)$, $h \in W$, $F_\beta h \in W$. To see this, fix $\beta \in (0,1)$, $h \in W$. We only need to consider the case $h > 0$. That the root $F_\beta h = y^*(s; h)$ is measurable follows from Filippov's implicit function theorem (e.g., Aliprantis and Border (2006), Theorem 18.17) noting y^* is a function. Additionally, under our assumptions, it can be show that there exists a strictly positive "lower bound" (or "subsolution") for F_β, denoted by $h_0 \in W$, such that for all $s \in S^* = (0, k_{\max}] \times Z$, $0 < h_0(s) < F_\beta h_0(s)$ (e.g., see McGovern et al. (2013), Proposition 2).

Finally, F_β is order continuous on $[h_0, w] \subset W$ where $[h_0, w]$ is sigma sub-complete in its relative partial order induced by W. To see this, first note $F_\beta h$ monotone increasing in (β, h): let $\beta_1 \leq \beta_2$, and $0 \leq h_1 \leq h_2 \leq w$. As for all (y, s), $Z_{\beta_1}(y; s, h_1) \leq Z_{\beta_2}(y; s, h_2)$ when $h_1 > 0$ (else, $F_\beta h_1 = 0 \leq F_\beta h_2$). As $Z_\beta(y; s, h)$ is strictly increasing in y, we have $F_{\beta_1} h_1 \leq F_{\beta_2} h_2$. So $F_\beta h$ is monotone increasing

[22] For the idd shock case, the space of candidate recursive equilibria can be taken to be additionally monotone increasing in z, and either upper or lower-semicontinuous in k. In such a case, the domain W can be made into a complete lattice. Unfortunately, on that space, the analogous operator F_β is not order continuous.

jointly in (β, h).

To show the operator F_β is then order continuous on $[h_0, w]$ for each $\beta \in (0,1)$, let $\{h_n\} \subset [h_0, w] \subset W$ be a countable chain. As $[h_0, w] \subset W$ is sigma subcomplete in W, $\sup\{h_n\} \uparrow h \in [h_0, w] \subset W$. As for each $\beta \in (0,1)$, F_β is monotone increasing on $[h_0, w] \subset W$, $\{F_\beta h_n\}$ is a countable chain in $[h_0, w]$. As F_β transforms $[h_0, w]$, by sigma-subcompleteness of $[h_0, w]$, $\sup\{F_\beta h_n\} \in [h_0, w]$. As the state space $[\underline{k}, \bar{k}] \times Z$ is compact, and the integrands in $Z_\beta(y; s, h)$ are continuous and bounded, the stochastic process π is Feller, by dominated convergence, we have $\sup\{F_\beta h_n\} \uparrow F_\beta h \in [h_0, w]$. Finally, as for each n, we have

$$Z_\beta(F_\beta h_n; s, h_n) = u'(w(s) - F_\beta h_n)$$
$$-\beta \int_Z v'(r(h_n(s), z') F_\beta h_n) r(F_\beta h_n, z') \pi(z, dz') = 0$$

as $Z_\beta(y; s, h_n)$ is continuous in (y, s, h_n) pointwise under our assumptions, for each $\beta \in (0,1)$, we conclude we have

$$F_\beta\left(\sup_n h_n\right) = \sup_n F_\beta h_n$$

As similar argument shows F_β on $[h_0, w]$ is inf-preserving. Hence, $F_\beta h$ is order continuous. We conclude applying Theorem 2 to the operator F_β. □

We conclude with three remarks about Proposition 5. First, Theorem 2 does not guarantee existence of RE. To see that the set of RE is non-empty in the analyzed economy, observe that $(A, \leq)$ is a sigma-complete lattice and $F_\beta : A \to A$ is an order continuous self map, hence, there exists an RE by an application of a Tarski-Kantorovich fixed point theorem.[23]

Second, in the interpretation, this proposition says that even in the presence of a possible continuum of RE in $(A, \leq)$, from any initial RE $h^*_{\beta_1} \in A$ of the less

[23] RE also exist (see Morand and Reffett (2007)) in subsets of W, where the elements of these subsets are additionally (i) lower or (ii) upper semicontinuous in k.

"patient" economy, the iterative process of computing the RE for the more patient economy is bounded below by $h^*_{\beta_1} \in A$.

Third, in dynamic stochastic equilibrium models with uncountable shocks the analyzed function space A (with the pointwise partial orders) is only sigma-complete. Hence, the results of Echenique (2002, 2004) and Sabarwal (2025b) can not be applied. We also obtain "order nearest" RE comparisons in the spirit of Sabarwal (2025b). Note, if the shocks are discrete, then the space $(A, \leq)$ is a complete lattice and by Tarski's theorem the set of RE on $(A, \leq)$ is a nonempty complete lattice for each β. The novelty of applying Theorem 2 in such settings is that it provides a monotone comparative statics of the long run (adaptive learning) bounds, and not only the comparative statics of extremal RE.

Applications of Theorem 2 can also be proposed in macroeconomic models with infinitely lived agents as well as with equilibrium indeterminacies (see for example Benhabib and Farmer (1994), Datta et al. (2018), and Pierri and Reffett (2026)). Even in nonconvex, nonoptimal dynamic economies, where the optimal household decisions are correspondences, Theorem 2 can be applied to obtain similar iterative monotone comparative statics result as these correspondences satisfy conditions of Theorem 2 under standard assumptions.[24]

A Appendix

Proof to Lemma 1 For $n = 0$ by definition and for $n > 1$ by induction: for any $1 \leq j \leq \gamma$ we have $\underline{a}^{n\gamma+j,\gamma} = f(\bigwedge\{\underline{a}^{n\gamma+k,\gamma} : k = 0, 1, \ldots, j-1\})$. Since $\bigwedge\{\underline{a}^{n\gamma+k,\gamma} : k = 0, 1, \ldots, j-1\} \leq \bigwedge\{\underline{a}^{n\gamma+k,\gamma} : k = 0, 1, \ldots, j-2\}$, we have $\underline{a}^{n\gamma+j,\gamma} = f(\bigwedge\{\underline{a}^{n\gamma+k,\gamma} : k = 0, 1, \ldots, j-1\}) \leq f(\bigwedge\{\underline{a}^{n\gamma+k,\gamma} : k = 0, 1, \ldots, j-2\}) = \underline{a}^{n\gamma+j-1,\gamma}$ for $j \geq 2$ by the monotonicity of f. Thus the sequence $(\underline{a}^{n\gamma+j,\gamma})_{j=1}^{\gamma}$ is decreasing.

[24] See Mirman et al. (2008) for examples of such economies and assumptions guaranteeing existence of RE.

As a consequence, $\underline{a}^{n\gamma+j,\gamma} = f(\underline{a}^{n\gamma+j-1,\gamma} \wedge \underline{a}^{n\gamma,\gamma})$ for any $1 \leq j \leq \gamma$. $\square$.

Lemma 2. *Let $\gamma \geq 2$ and consider $(\overline{a}^{k,\gamma})_{k=0}^{\infty}$, where $\overline{a}^{k+1,\gamma} = f(\bigvee\{\overline{a}^{k,\gamma}, ..., \overline{a}^{k-\gamma+1,\gamma}\})$ for some increasing mapping $f : A \to A$ and given $a^{0,\gamma} \in A$. Then:*

1. *for any $n \geq 0$, $j \in \{1, ..., \gamma - 1\}$ we have $\overline{a}^{n\gamma+j,\gamma} = f(\overline{a}^{n\gamma,\gamma} \vee \overline{a}^{n\gamma+j-1,\gamma}\})$,*

2. *for any $n \geq 1$ we have $\overline{a}^{n\gamma,\gamma} = f(\overline{a}^{(n-1)\gamma,\gamma} \vee \overline{a}^{n\gamma-1,\gamma})$.*

We skip its proof as it is analogous to that of Lemma 1.

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